ALLEN CAREER INSTITUTE
KOTA (RAJASTHAN)
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PHYSICS

JEE (MAIN+ADVANCED)

ENTHUSIAST COURSE

EXERCISE

Modern Physics-1

English Medium

EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

A. Photoelectric Effect

1. **Statement 1 :** Photoelectric effect establishes quantum nature of light.
and
Statement 2 : There is negligible time lag between photon collisions with the material and photoelectron emission irrespective of intensity of incident light. (Assume incident light is of frequency greater than threshold frequency of the material).
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

MP0043

2. **Statement-1 :** Work function of aluminum is 4.2 eV. If two photons each of energy 2.5 eV strikes on a piece of aluminum, the photo electric emission does not occur
Statement-2 : In photo electric effect a single photon interacts with a single electron and electron is emitted only if energy of each incident photon is greater then the work function.
(A) Statement-1 is True, Statement-2 is True, Statement-2 is a correct explanation for statement-1
(B) Statement-1 is True, Statement-2 is True, Statement-2 is NOT a correct explanation for Statement-1
(C) Statement-1 is True, Statement-2 is False
(D) Statement-1 is False, Statement-2 is True

MP0044

3. A photocell is illuminated by a small bright source placed 1 m away. When the same source of light is place $\frac{1}{2}$ m away, the number of electrons emitted by photocathode would- **[AIEEE - 2005]**

(A) decrease by a factor of 4
(B) increase by a factor of 4
(C) decrease by a factor of 2
(D) increase by a factor of 2

MP0045

4. Two monochromatic light sources, A and B, emit the same number of photons per second. The wavelength of A is $\lambda_A = 400$ nm, and that of B is $\lambda_B = 600$ nm. The power radiated by source B is
(A) equal to that of source A
(B) less than that of source A
(C) greater than that of source A
(D) cannot be compared to that from source A using the available data.

MP0046

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5. The energy flux of sunlight reaching the surface of the earth is 1.388×10^3 W/m^2. How many photons (nearly) per square metre are incident on the Earth per second? Assume that the photons in the sunlight have an average wavelength of 550 nm. **NCERT**

(A) 8×10^{21} (B) 4×10^{21} (C) 4×10^{38} (D) 8×10^{38}

MP0047

6. The threshold frequency for a certain metal is 3.3×10^{14} Hz. If light of frequency 8.2×10^{14} Hz is incident on the metal, the cutoff voltage for the photoelectric emission is. **NCERT**

(A) 8 V (B) 6 V (C) 2 V (D) 4 V

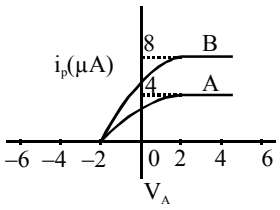
MP0048

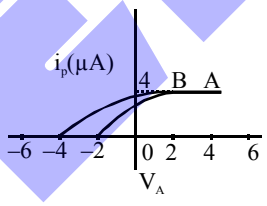
7. Light of frequency 7.21×10^{14} Hz is incident on a metal surface. Electrons with a maximum speed of 6.0×10^5 m/s are ejected from the surface. The threshold frequency for photoemission of electrons is **NCERT**

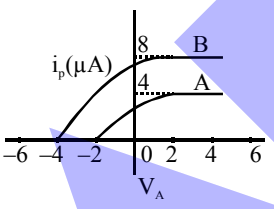
(A) 4.73×10^{14} Hz (B) 4.73×10^{10} Hz (C) 2.08×10^{10} Hz (D) None of these

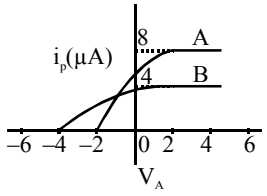
MP0049

8. In a photoelectric effect experiment, photons of energy 5 eV are incident on the photo-cathode of work function 3 eV. For photon intensity $I_A = 10^{15}$ m^{-2}s^{-1}, saturation current of 4.0 μA is obtained. Sketch of the variation of photocurrent i_p against the anode voltage V_a for photon intensity I_A (curve A) and $I_B = 2 \times 10^{15}$ m^{-2} s^{-1} (curve B) will be :

(A) 

(B) 

(C) 

(D) 

MP0050

9. **Statement-1 :** When ultraviolet light is incident on a photocell, its stopping potential is V_0 and the maximum kinetic energy of the photoelectrons is K_{max}. When the ultraviolet light is replaced by infrared light, both V_0 and K_{max} increase.

Statement-2 : Photoelectrons are emitted with speeds ranging from zero to a maximum value.

(A) Statement–1 is true, Statement–2 is false

(B) Statement–1 is true, Statement–2 is true; Statement–2 is the correct explanation of Statement–1

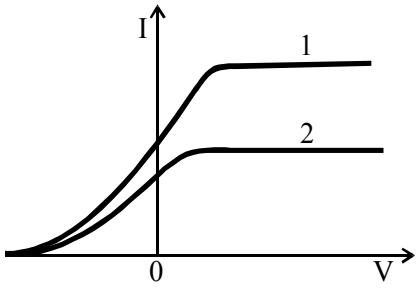
(C) Statement–1 is true, Statement–2 is true; Statement–2 is not the correct explanation of Statement–1

(D) Statement–1 is false, Statement–2 is true

MP0051

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10. A photocathode can be illuminated by the light from two sources, each of which emits monochromatic radiation. The sources are positioned at equal distances from the photocathode. The dependence of the photocurrent on the voltage between the cathode and the anode is depicted by curve 1 for one source and by curve 2 for the other. In what respect do these **sources** differ ?



(A) Highest frequency photon
(B) Number of photons emmited per second
(C) Number of photoelectrons emmited per second
(D) None of these

MP0052

11. A radiation of energy E falls normally on a perfectly reflecting surface. The momentum transferred to the surface is- **[AIEEE - 2004]**

(A) E/c (B) 2E/c (C) Ec (D) E/c^2

MP0053

12. The threshold frequency for a metallic surface corresponds to an energy of 6.2 eV and the stopping potential for a radiation incident on this surface is 5V. The incident radiation lies in- **[AIEEE - 2006]**

(A) ultra-violet region (B) infra-red region
(C) visible region (D) X-ray region

MP0054

13. The time taken by a photoelectron to come out after the photon strikes is approximately- **[AIEEE - 2006]**

(A) 10^{-4} s (B) 10^{-10} s (C) 10^{-16} (D) 10^{-1} s

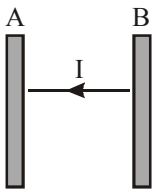
MP0055

14. Photon of frequency ν has a momentum associated with it. If c is the velocity of light, the momentum is- **[AIEEE - 2007]**

(A) ν/c (B) hνc (C) $h\nu/c^2$ (D) hν/c

MP0057

15. A parallel beam of light of intensity I is incident normally on a plane surface A which absorbs 50% of the incident light. The reflected light falls on B which is perfect reflector, the light reflected by B is again partly reflected and partly absorbed and this process continues. For all absorption by A, asborption coefficient is 0.5. The pressure experienced by A due to light is :-



(A) $\frac{1.5\,I}{c}$ (B) $\frac{I}{c}$ (C) $\frac{3I}{2c}$ (D) $\frac{3I}{c}$

MP0065

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B. De-Broglie hypothesis

16. **Statement-1 :** If an electron has the same wavelength as a photon, they have the same energy.
Statement-2 : by debroglie hypothesis, $p = h/\lambda$ for both the electron and the photon.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

MP0058

17. A particle of mass 4m at rest decays into two particles of masses m and 3m having non-zero velocities. The ratio of the de-Broglie wavelengths of the particles 1 and 2 is

(A) $\frac{1}{2}$ (B) $\frac{1}{4}$ (C) 2 (D) 1

MP0059

18. A free particle with initial kinetic energy E and de–broglie wavelength λ enters a region in which it has potential energy U. What is the particle's new de–Broglie wavelength?
(A) $\lambda\,(1 - U/E)^{-1/2}$ (B) $\lambda\,(1 - U/E)$
(C) $\lambda\,(1 - E/U)^{-1}$ (D) $\lambda\,(1 + U/E)^{1/2}$

MP0060

19. Proton, deutron and α particles are accelerated through the same potential difference. Then the ratio of their de-Broglie wavelength as

(A) $1:\sqrt{2}:1$ (B) 1 : 1 : 1
(C) $1:2:2\sqrt{2}$ (D) $2\sqrt{2}:2:1$

MP0061

20. **Statement-1 :** An electron and a proton are accelerated through the same potential difference. The de-Broglie wavelength associated with the electron is longer.

Statement-2 : De-Broglie wavelength associated with a moving particle is $\lambda = \frac{h}{p}$ where, p is the linear momentum and both have same KE.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

MP0062

21. An α-particle of energy 5 MeV is scattered through 180° by a fixed uranium nucleus. The distance of the closest approach is of the order of- **[AIEEE-2004]**
(A) 1 Å (B) 10^{-10} cm (C) 10^{-12} cm (D) 10^{-15} cm

MP0063

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22. After absorbing a slowly moving neutron of mass m_N (momentum ~0) a nucleus of mass M breaks into two nuclei of masses m_1 and $3m_1 (4m_1 = M + m_N)$, respectively. If the de Broglie wavelength of the nucleus with mass m_1 is λ, then de-Broglie wavelength of the other nucleus will be:-

(A) 9λ (B) 3λ (C) $\frac{\lambda}{3}$ (D) λ

MP0064

23. A beam of electrons with de-broglie wavelength of 10^{-4} m pass through a slit 10^{-3} m wide. Calculate the angular spread introduced because of diffraction by slit.

(A) $\frac{9°}{\pi}$ (B) $\frac{18°}{\pi}$ (C) $\frac{36°}{\pi}$ (D) $\frac{4.5°}{\pi}$

WO0098

24. This question has Statement-1 and statement-2. Of the four choices given after the Statements, choose the one that best describes the two statements. **[AIEEE - 2012]**

Statement-1 : Davisson - Germer experiment established the wave nature of electrons.

Statement-2: If electrons have wave nature, they can interfere and show diffraction.

(A) Statement-1 is true, Statement-2 is true and Statement-2 is the correct explanation for Statement-1.

(B) Statement-1 is true, Statement-2 is true and Statement-2 is not the correct explanation of Statement-I

(C) Statement-1 is false, Statement-2 is true.

(D) Statement-1 is true, Statement-2 is false

WO0119

C. Atomic structure

25. The de-Broglie wavelength of an electron in the first Bohr orbit is

(A) equal to the circumference of first orbit (B) equal to $\frac{1}{2}$ × (circumference of first orbit)

(C) equal to $\frac{1}{4}$ × (circumference of first orbit) (D) equal to $\frac{3}{4}$ × (circumference of first orbit)

MP0068

26. **Statement–1** : When light is passed through a sample of hydrogen atoms in ground state, then wavelengths of absorption lines are same as wavelengths of lines of Lyman series in emission spectrum.

and

Statement–2 : In ground state hydrogen atom will absorb only those radiation which will excite to higher energy level.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

MP0069

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27. **Statement-1 :** In a laboratory experiment, on emission from atomic hydrogen in a discharge tube, only a small number of lines are observed whereas a large number of lines are present in the hydrogen spectrum of a star.

Statement-2 : The temperature of discharge tube is much smaller than that of the star.

(A) Statement-1 is True, Statement-2 is True, Statement-2 is a correct explanation for statement-1

(B) Statement-1 is True, Statement-2 is True, Statement-2 is NOT a correct explanation for Statement-1

(C) Statement-1 is True, Statement-2 is False

(D) Statement-1 is False, Statement-2 is True

MP0070

28. If the binding energy of the electron in a hydrogen atom is 13.6 eV, the energy required to remove the electron from the first excited state of Li^{2+} is- **[AIEEE - 2003]**

(A) 30.6 eV (B) 13.6 eV (C) 3.4 eV (D) 122.4 eV

MP0071

29. A neutron having kinetic energy 13.6 eV collides with a hydrogen atom in ground state at rest. Assume that the mass of neutron and hydrogen atoms are same and the neutron does not leave its line of motion. Then which of the following is the possible KE of the neutron after the collision?

(A) zero (B) 3.4 eV (C) 1.5 eV (D) 6.8 eV.

MP0072

30. In the spectrum of single ionised helium, the wavelength of a line observed is almost the same as the first line of Balmer series of hydrogen. It is due to transition of electron from

(A) $n_1 = 6$ to $n_2 = 4$ (B) $n_1 = 5$ to $n_2 = 3$

(C) $n_1 = 4$ to $n_2 = 2$ (D) $n_1 = 3$ to $n_2 = 2$

MP0073

31. Light coming from a discharge tube filled with hydrogen falls on the cathode of the photoelectric cell. The work function of the surface of cathode is 4eV. Which one of the following values of the anode voltage (in Volts) with respect to the cathode will likely to make the photo current zero.

(A) – 4 (B) – 6 (C) – 8 (D) – 10

MP0074

32. According to Bohr model, magnetic field at the centre (at the nucleus) of a hydrogen atom due to the motion of the electron in n^{th} orbit is proportional to

(A) $1/n^3$ (B) $1/n^5$ (C) n^5 (D) n^3

MP0075

33. A photon of 10.2 eV energy collides with a hydrogen atom in ground state inelastically. After few microseconds one more photon of energy 15 eV collides with the same hydrogen atom.Then what can be detected by a suitable detector. **[JEE' 2005 (Scr)]**

(A) one photon of 10.2 eV and an electron of energy 1.4 eV

(B) 2 photons of energy 10.2 eV

(C) 2 photons of energy 3.4 eV

(D) 1 photon of 3.4 eV and one electron of 1.4 eV

MP0076

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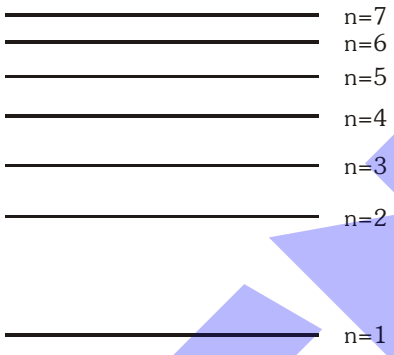
34. Two hydrogen atoms are in excited state with electrons residing in n = 2. First one is moving towards left and emits a photon of energy E_1 towards right. Second one is moving towards left with same speed and emits a photon of energy E_2 towards left. Taking recoil of nucleus into account during emission process

(A) $E_1 > E_2$ (B) $E_1 < E_2$

(C) $E_1 = E_2$ (D) information insufficient

MP0077

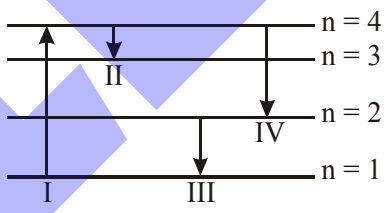
35. The maximum number of emission lines for atomic hydrogen that you would expect to see with naked eye if the only electronic levels involved are those shown in the figure, is



(A) 6 (B) 5 (C) 21 (D) ∞

MP0078

36. The diagram shows the energy levels for an electron in a certain atom. Which transition shown represents the emission of a photon with maximum wavelength?



(A) III (B) IV (C) I (D) II

MP0079

37. The largest wavelength in the ultraviolet region of the hydrogen spectrum is 122 nm. The smallest wavelength in the infrared region of the hydrogen spectrum (to the nearest integer) is

[JEE 2007]

(A) 802 nm (B) 823 nm (C) 1882 nm (D) 1648 nm

MP0080

38. Which of the following transitions in hydrogen atoms emit photons of highest frequency ?

[AIEEE - 2007]

(A) n = 2 to n = 6 (B) n = 6 to n = 2

(C) n = 2 to n = 1 (D) n = 1 to n = 2

MP0093

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D. X-Rays

39. **Statement-1 :** If the accelerating potential in an X-ray tube is increased, the wavelengths of the characteristic X-rays do not change. **[JEE 2007]**

because

Statement-2 : When an electron beam strikes the target in an X-ray tube, part of the kinetic energy is converted into X-ray energy

(A) Statement-1 is True, Statement-2 is True; Statement-2 **is** a correct explanation for Statement-1

(B) Statement-1 is True, Statement-2 is True; Statement-2 **is NOT** a correct explanation for Statement-1

(C) Statement-1 is True, Statement-2 is False

(D) Statement-1 is False, Statement-2 is True

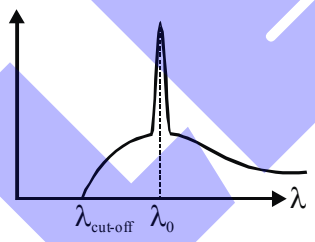
MP0066

40. Electrons with de-Broglie wavelength λ fall on the target in an X-ray tube. The cut-off wavelength of the emitted X-rays is **[JEE 2007]**

(A) $\lambda_0 = \frac{2mc\lambda^2}{h}$ (B) $\lambda_0 = \frac{2h}{mc}$ (C) $\lambda_0 = \frac{2m^2c^2\lambda^3}{h^2}$ (D) $\lambda_0 = \lambda$

MP0067

41. **Assertion (A) :** Variation of intensity of X-rays is plotted against λ. On increasing the accelerating potential, $(\lambda_0 - \lambda_{cut\text{-}off})$ increases.



Reasoning (R) : $\lambda_{cut\text{-}off}$ will decrease but λ_0 will be same, as wavelength of characteristic X-rays is independent of the accelerating potential.

Choose the correct statement from the following.

(A) A is correct and R is the correct explanation of A.

(B) Both A and R are correct but R is not the correct explanation of A.

(C) A is correct but R is wrong.

(D) Both A and R are wrong.

MP0081

42. An X-ray tube is run at 50 kV. The current flowing in it is 20mA. The power of the tube is :

(A) 1000 W (B) 200 W (C) 20000 W (D) 20 W

MP0082

43. Choose the **INCORRECT** statement.

(A) Cut-off wavelength of X-rays is independent of filament voltage.

(B) Presence of K_α-line in X-ray spectrum means that L-series will also be present.

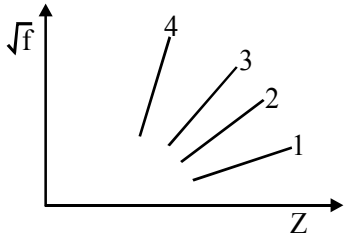
(C) Increase in filament current increases intensity of X-ray.

(D) Presence of L-series in X-ray spectrum means that K-series will also be present.

MP0083

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44. The given graph shows the variation of $\sqrt{f}$ vs Z for characteristics X-rays. Lines 1, 2, 3, 4 shown in the graph corresponds to any one of k_α, k_β, L_α, L_β. Then L_β is represent by :-



(A) line 1 (B) line 2 (C) line 3 (D) line 4

MP0084

45. The X-ray beam coming from an X-ray tube will be
(A) monochromatic
(B) having all wavelengths smaller than a certain maximum wavelength
(C) having all wavelengths larger than a certain minimum wavelength
(D) having all wavelengths lying between a minimum and a maximum wavelength

MP0085

46. E_1 is energy of k_α photon of aluminium, E_2 is energy of k_β photon of aluminium and E_3 is energy of k_α photon from sodium, then the correct order of energies is given
(A) $E_1 > E_2 > E_3$ (B) $E_3 > E_2 > E_1$ (C) $E_3 > E_1 > E_2$ (D) $E_2 > E_1 > E_3$

MP0086

47. The K,L and M energy levels of platinum lie roughly at 78, 12 and 3 keV respectively. The ratio of wavelength of K_α line to that of K_β line in X-ray spectrum is-

(A) $\frac{22}{3}$ (B) $\frac{3}{22}$ (C) $\frac{22}{25}$ (D) $\frac{25}{22}$

MP0087

48. Mosley law relates :
(A) Frequency of emitted X-ray with applied voltage
(B) Wavelength and intensity of X-ray.
(C) Frequency of emitted X-ray with atomic number
(D) Wavelength and angle of scattering.

MP0089

49. The wavelength of K_α X-ray of an element having atomic number Z = 11 is λ . The wavelength of K_α X-ray of another element of atomic number Z' is 4λ. Then Z' is **[JEE' 2005 (Scr)]**
(A) 11 (B) 44 (C) 6 (D) 4

MP0091

50. Characteristic X-ray
(A) Have only discrete wavelength which are characteristic of the target.
(B) Have all the possible wavelength.
(C) Are characteristic of speed of projectile electrons.
(D) None

MP0092

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EXERCISE (O-2)

SINGLE CORRECT TYPE QUESTIONS

A. Photoelectric Effect

1. The maximum kinetic energy of photo-electron liberated from the surface of lithium (work function = 2.35 eV) by electromagnetic radiation whose electric component varies with time as $E = a[1 + \cos(2\pi f_1 t)]\cos(2\pi f_2 t)$ where 'a' is a constant, $f_1 = 3.6 \times 10^{15}$ Hz and $f_2 = 1.2 \times 10^{15}$ Hz is [Take : $h = 6.6 \times 10^{-34}$ J-s]

(A) 2. 6 eV (B) 7.55 eV (C) 12.5 eV (D) 17.45 eV

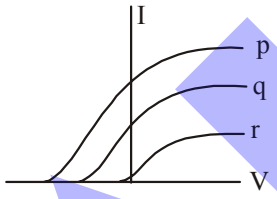
MP0119

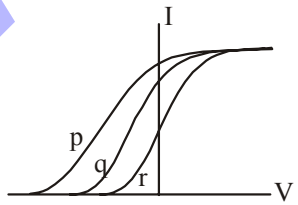
2. A photocell in the saturation mode is irradiated by light of wavelength $\lambda = 6600$ Å. The corresponding spectral sensitivity of the cell is $s_\lambda = 4.8$ mA/W. Find the yield of photoelectrons, i.e. the number of photoelectrons produced by each incident photon. [Take : $h = 6.6 \times 10^{-34}$ J-s]

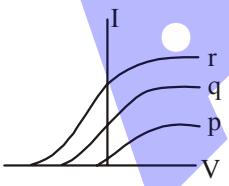
(A) 9×10^{-2} (B) 9×10^{-4} (C) 9×10^{-3} (D) 9

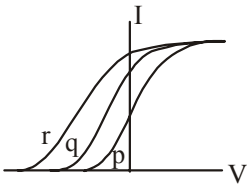
MP0120

3. Photoelectric effect experiments are performed using three different metal plates p, q and r having work functions $\phi_p = 2.0$ eV, $\phi_q = 2.5$ eV and $\phi_r = 3.0$ eV, respectively. A light beam containing wavelengths of 550 nm, 450 nm and 350 nm with equal intensities illuminates each of the plates. The correct I-V graph for the experiment is : **[JEE-2009]**

(A) 

(B) 

(C) 

(D) 

MP0167

4. When a monochromatic point-source of light is at a distance of 0.2 m from a small photoelectric cell, the stopping potential and the saturation current are respectively 0.6 volt and 18.0 mA. If the same source is placed 0.6 m away from the photoelectric cell, then :-

(A) The stopping potential will be 0.2 volt (B) The stopping potential will be 1.8 volt

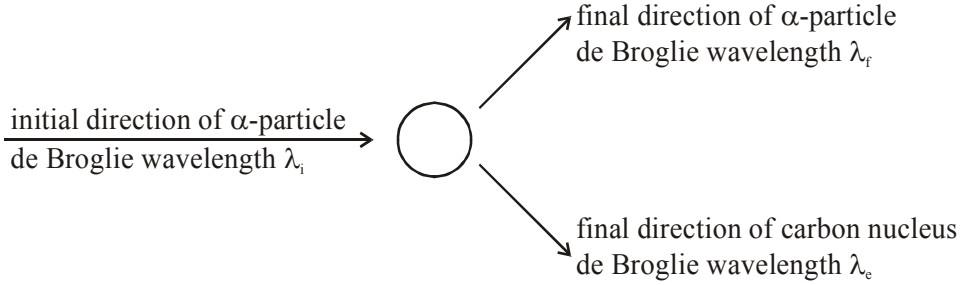
(C) The saturation current will be 6.0 mA (D) The saturation current will be 2.0 mA

MP0135

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B. De-Broglie hypothesis

5. An α-particle having a de Broglie wavelength λ_i collides with a stationary carbon nucleus. The α-particle moves off in a different direction as shown below.



After the collision, the de Broglie wavelength of the α-particle and the carbon nucleus are λ_f and λ_e respectively. Which of the following relations about de Broglie wavelength is correct ?

(A) $\lambda_i < \lambda_f$ (B) $\lambda_i > \lambda_f$ (C) $\lambda_f = \lambda_e$ (D) $\lambda_i = \lambda_e$

MP0121

C. Atomic structure

6. Apply Bohr's atomic model to a lithium atom. Assuming that its two K-shell electrons are too close to nucleus such that nucleus and K-shell electron act as a nucleus of effective positive charge equivalent to electron. The ionization energy of its outermost electron is:-

(A) 30.6 eV (B) 3.4 eV (C) 32.4 eV (D) 13.6 eV

MP0123

7. The attractive potential for an atom is given by $v = v_0 ln(r/r_0)$, v_0 and r_0 are constant and r is the radius of the orbit. The radius r of the nth Bohr's orbit depends upon principal quantum number n as:

[JEE' 2003 (Scr)]

(A) $r \propto n$ (B) $r \propto 1/n^2$ (C) $r \propto n^2$ (D) $r \propto 1/n$

MP0124

8. A force of attraction between the positively charged nucleus and the negatively charged electron in the hydrogen atom is given by $F = \frac{ke^2}{r^2}$ where k is the constant. The electron, initially moving in a circle of radius R_1 about the nucleus, jumps suddenly into a circular orbit of radius R_2. The total energy of the atom decreased in this process is :-

(A) $ke^2\left[\frac{1}{R_1}+\frac{1}{R_2}\right]$ (B) $\frac{ke^2}{2}\left[\frac{1}{R_2}-\frac{1}{R_1}\right]$ (C) $ke^2\left[\frac{R_1R_2}{R_2-R_1}\right]$ (D) $ke^2\left[\frac{R_1R_2}{R_1+R_2}\right]$

MP0125

9. A stationary hydrogen atom of mass M emits a photon corresponding to the longest wavelength of Balmer series. The recoil velocity acquired by the atom is (R = Rydberg constant and h = plank's constant)

(A) $\frac{Rh}{M}$ (B) $\frac{Rh}{4M}$ (C) $\frac{3}{4}\frac{Rh}{M}$ (D) $\frac{5}{36}\frac{Rh}{M}$

MP0126

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10. Let the potential energy of a hydrogen atom in the ground state be zero. Then its energy in the first excited state will be :

(A) 10.2eV (B) 13.6eV (C) 23.8eV (D) 27.2 eV

MP0127

11. Hydrogen atoms in ground state are excited by monochromatic radiation of wavelength 975 Å. The number of liens in the resulting spectrum will be :

(A) 3 (B) 4 (C) 6 (D) 10

MP0128

12. The relation between λ_1 : wavelength of series limit of lyman λ_2 : the wavelength of the series limit of Balmer series and λ_3 : the wavelength of first line of lyman series is

(A) $\lambda_1 = \lambda_2 + \lambda_3$ (B) $\lambda_3 = \lambda_1 + \lambda_2$ (C) $\lambda_2 = \lambda_3 - \lambda_1$ (D) none

MP0129

13. A hydrogen like gas atoms absorb radiations of wavelength λ_0 and consequently emit radiations of 6 difference wavelengths of which, three wavelengths are shorter than λ_0. Choose the correct alternative(s).

(A) The final excited state of the atoms is n = 3.
(B) The final excited state of the atoms is n = 4.
(C) The initial state of the atoms is n = 1.
(D) The initial state of the atoms is n = 3.

MP0130

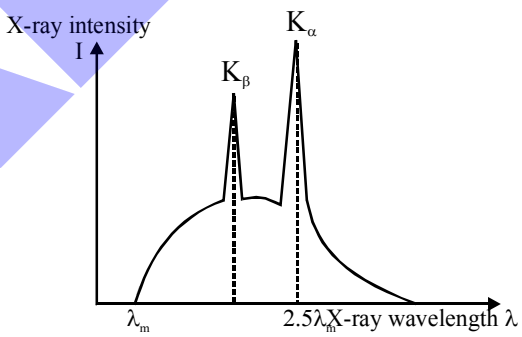
14. According to the Bohr theory of the hydrogen atom, electron starting in the 4th energy level and eventually ending in the ground state could produce a total of how many lines in the hydrogen spectra?

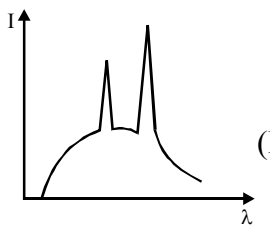
(A) 7 (B) 6 (C) 5 (D) 4
(E) 3

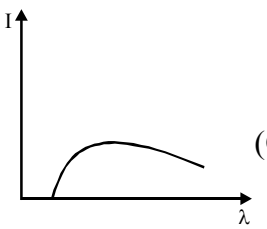
MP0131

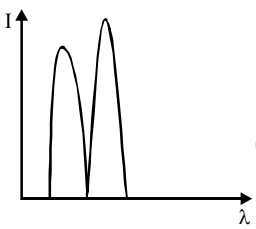
D. X-Rays

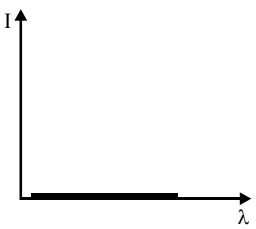
15. When an electron accelerated by potential difference U is bombarded on a specific metal the emitted X-ray spectrum obtained is shown in adjoining graph. If the potential difference is reduced to U/3, the correct spectrum is :-



(A) 

(B) 

(C) 

(D) 

MP0132

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16. In an X-ray experiment target is made up of copper (Z = 29) having some impurity. The K_α line of copper have wavelength λ_0. It was observed that another K_α line due to impurity have wavelength $\frac{784}{625}\lambda_0$. The atomic number of the impurity element is

(A) 22 (B) 23 (C) 24 (D) 26

MP0133

MULTIPLE CORRECT TYPE QUESTIONS

A. Photoelectric Effect

17. In a photoelectric effect experiment, if f is the frequency of radiations incident on the metal surface and I is the intensity of the incident radiations, then choose the correct statement(s).

(A) If f is increased keeping I and work function constant then maximum kinetic energy of photoelectron increases.

(B) If distance between cathode and anode is increased stopping potential increases

(C) If I is increased keeping f and work function constant then stopping potential remains same and saturation current increases.

(D) Work function is decreased keeping f and I constant then stopping potential increases

MP0134

18. The collector of the photocell (in photoelectric experiment) is made of tungsten while the emitter is of Platinum having work function of 10 eV. Monochromatic radiation of wavelength 124 Å & power 100 watt is incident on emitter which emits photo electrons with a quantum efficiency of 1%. The accelerating voltage across the photocell is of 10,000 volts (Use : hc = 12400eV Å)

(A) The power supplied by the accelerating voltage source is 100 watt

(B) The minimum wavelength of radiation coming from the tungsten target (collector) is 1.23 Å

(C) The power supplied by the accelerating voltage source is 10 watt

(D) The minimum wavelength of radiation coming from the tungsten target (collector) is 2.23 Å

MP0136

19. Light of wavelength λ_1 & λ_2 are falling on two metal surface A & B. For wavelength λ_1 electron ejected from both the surfaces and for wavelength λ_2 electron ejected from only surface B. On the basis of these facts which one of the following is a false statement

(A) more energy is required for ejection of electron from metal 'A'

(B) $\lambda_1 > \lambda_2$

(C) threshold wavelength for A is greater than B

(D) energy of electron ejected from metal A will be greater than electron ejected from metal B for wavelength λ

MP0137

20. The accelerating potential (V) applied between a photocathode and the respective anode is such that the fastest photoelectron can fly only one fourth of the distance between the cathode and the anode. If the distance between photocathode and anode is reduced to $(1/4)^{th}$ of the original value while maintaining the accelerating potential constant, then

(A) The fastest electron will reach the anode

(B) The fastest electron will reach upto one fourth of the new distance between the cathode and the anode

(C) Kinetic energy of the fastest photoelectron emitted from photocathode will not change due to change in plate separation (keeping V constant)

(D) Kinetic energy of the fastest photoelectron will increase due to decrease in plate separation (keeping V constant)

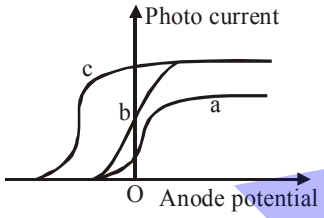
MP0138

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21. Photoelectric effect supports quantum nature of light because
(A) there is minimum frequency of light below which no photoelectrons are emitted
(B) the maximum kinetic energy of photo-electrons depends only on the frequency of light and not on its intensity
(C) even when the metal surface is faintly illuminated, the photoelectrons leave the surface immediately
(D) electric charge of photo-electrons is quantized

MP0094

22. The figure shows the variation of photo current with anode potential for a photosensitive surface for three different radiations. Let I_a, I_b and I_c be the intensities and f_a, f_b and f_c be the frequencies for the curves a, b and c respectively. Choose correct options



(A) $f_a = f_b$ (B) $I_a < I_b$ (C) $f_c < f_b$ (D) $I_c > I_b$

MP0095

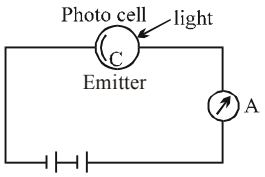
23. In photoelectric effect, stopping potential depends on
(A) frequency of the incident light
(B) intensity of the incident light by varies source distance
(C) emitter's properties
(D) frequency and intensity of the incident light

MP0096

24. Which of the following phenomena can be explained only on the basis of quantum theory of light?
(A) Energy spectrum of black body radiation (B) Atomic spectra
(C) Photoelectric effect (D) Doppler effect

MP0097

25. The figure shows a photo cell circuit. The cathode of the photo cell is illuminated by a monochromatic light. If the intensity is kept constant and the frequency of the incident light is increased, then the



(A) photo electric current in the circuit increases
(B) photo electric current in the circuit decreases
(C) maximum kinetic energy of the photo electrons increases
(D) photo electric current in the circuit can be reduced to zero, when the polarity of the terminals is reversed

MP0098

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26. In a photoelectric effect, electrons are emitted
(A) at a rate that is proportional to the square of the amplitude of the incident radiation.
(B) with a maximum velocity proportional to the frequency of the incident radiation.
(C) at a rate that is independent of the emitter.
(D) only if the frequency of the incident radiation is above a certain threshold value
(E) only if the temperature of the emitter is high

MP0099

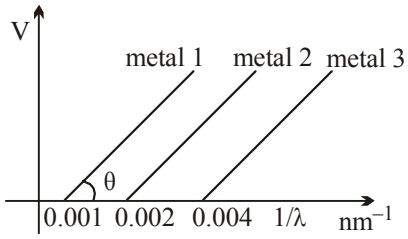
27. A small plate of area 1 cm^2 is placed at a distance of $\frac{1}{\sqrt{\pi}}$ m from an isotropic point source emitting light of frequency $\frac{1}{6.63}\times10^{14}$ Hz, at a power of 2.00 mW. Assume the plate to be normal to the incident photons. [where $h = 6.63 \times 10^{-34}$ J-s]. Select **CORRECT** alternative(s)
(A) Energy possessed by each photon is 10^{-20} J
(B) Photon emission rate is 2×10^{17} s^{-1}
(C) The fraction of area of beam intercepted by the plate is $\frac{1}{4}\times10^{-4}$
(D) The rate of photons striking the plate is 5×10^{12} per second.

MP0100

28. A metallic sphere of radius r remote from all other bodies is irradiated with a radiation of wavelength λ which is capable of causing photoelectric effect.
(A) the maximum potential gained by the sphere will be independent of its radius
(B) the net positive charge appearing on the sphere after a long time will depend on the radius of the sphere
(C) the maximum kinetic energy of the electrons emanating from the sphere will keep on declining with time
(D) the kinetic energy of the most energetic electrons emanating from the sphere initially will be independent of the radius of the sphere.

MP0101

29. The graph between $1/\lambda$ and stopping potential (V) of three metals having work functions ϕ_1, ϕ_2 and ϕ_3 in an experiment of photo-electric effect is plotted as shown in the figure. Which of the following statement(s) is/are correct? [Here λ is the wavelength of the incident ray]. **[JEE 2006]**



(A) Ratio of work functions $\phi_1 : \phi_2 : \phi_3 = 1 : 2 : 4$
(B) Ratio of work functions $\phi_1 : \phi_2 : \phi_3 = 4 : 2 : 1$
(C) tan θ is directly proportional to hc/e, where h is Planck's constant and c is the speed of light
(D) The violet colour light can eject photoelectrons from metals 2 and 3.

MP0113

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B. De-Broglie hypothesis

30. Two electrons are moving with the same speed v. One electron enters a region of uniform electric field while the other enters a region of uniform magnetic field, then after sometime if the de–Broglie wavelengths of the two are λ_1 and λ_2, then select the possible option(s) :-

(A) $\lambda_1 = \lambda_2$ (B) $\lambda_1 > \lambda_2$ (C) $\lambda_1 < \lambda_2$ (D) $\lambda_1 > \lambda_2$ or $\lambda_1 < \lambda_2$

MP0102

31. A particle moves in a closed orbit around the origin, due to a central force which is directed towards the origin. The de Broglie wavelength of the particle varies cyclically between two values λ_1, λ_2 with $\lambda_1 > \lambda_2$. Which of the following statements is/are true ?

(A) The particle could be moving in a circular orbit with origin as centre

(B) The particle could be moving in an elliptic orbit with origin as its focus.

(C) When the de Broglie wave length is λ_1, the particle is nearer the origin than when its value is λ_2.

(D) When the de Broglie wavelength is λ_2, the particle is nearer the origin than when its value is λ_1.

MP0103

32. A beam of 2000 eV electrons are incident normally on the surface of crystal whose inter atomic separation is 0.11 nm. The mass of the electron can be taken as 9×10^{-31} kg. At what angle to the normal can we observe a diffraction maxima.

(A) $\sin^{-1}\left(\frac{1}{4}\right)$ (B) $2\cos^{-1}\left(\frac{1}{4}\right)$ (C) $\sin^{-1}\left(\frac{1}{2}\right)$ (D) $2\cos^{-1}\left(\frac{1}{2}\right)$

WO0068

C. Atomic structure

33. Choose the correct statement(s) for hydrogen and deuterium atoms (considering motion of nucleus)

(A) The radius of first Bohr orbit of deuterium is less than that of hydrogen

(B) The speed of electron in the first Bohr orbit of deuterium is more than that of hydrogen.

(C) The wavelength of first Balmer line of deuterium is more than that of hydrogen

(D) The angular momentum of electron in the first Bohr orbit of deuterium is more than that of hydrogen.

MP0122

34. For the electron in the nth orbit of hydrogen atom. Under the assumption of the Bohr's atomic model choose the **CORRECT** option(s) (Here n is the principal quantum number) :-

(A) Frequency of the electron is inversely proportional to n^3

(B) The magnitude of potential energy of the electron in an orbit is greater than its kinetic energy

(C) Magnetic induction at the nucleus produced due to the motion of electron in the nth orbit is proportional to n^5

(D) Magnetic moment produced due to the motion of electron in the nth orbit is proportional to n

MP0139

35. Energy liberated in the de-excitation of hydrogen atom from 3rd level to 1st level falls on a photo-cathode. Later when the same photocathode is exposed to a spectrum of some unknown hydrogen like gas, excited to 2nd energy level, it is found that the de-Broglie wavelength of the fastest photoelectrons, now ejected has decreased by a factor of 3. For this new gas, difference of energies of 2nd Lyman line and 1st Balmer line is found to be 3 times the ionization potential of the hydrogen atom. Select the correct statement(s) :

(A) The gas is lithium

(B) The gas is helium

(C) The work function of photocathode is 8.5 eV

(D) The work function of photocathode is 5.5 eV

MP0140

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36. An electron of the kinetic energy 10eV collides with a hydrogen atom in 1st excited state. Assuming loss of kinetic energy in the collision to be quantized, the collision :
(A) may be perfectly inelastic
(B) may be inelastic
(C) may be elastic
(D) must be inelastic

MP0141

37. According to Bohr's theory of hydrogen atom, for the electron in the nth permissible orbit,

(A) linear momentum $\propto \frac{1}{n}$
(B) radius of orbit $\propto n$
(C) kinetic energy $\propto \frac{1}{n^2}$
(D) angular momentum $\propto n$

MP0104

38. The magnitude of angular momentum, orbit radius and frequency of an electron in hydrogen atom corresponding to the quantum number n are L, r and f respectively, then according to Bohr's theory of hydrogen atom.

(A) frL is constant for all orbits
(B) $Lf \propto \frac{1}{n^2}$
(C) $fr \propto \frac{1}{n}$
(D) $Lr \propto \frac{1}{n^3}$

MP0105

39. A particular hydrogen like atom has its ground state binding "energy 122.4eV. Its is in ground state. Then:
(A) Its atomic number is 3
(B) An electron of 90eV can excite it.
(C) An electron of kinetic energy nearly 91.8eV can be brought to almost rest by this atom.
(D) An electron of kinetic energy 2.6eV may emerge from the atom when electron of kinetic energy 125eV collides with this atom.

MP0106

40. A beam of ultraviolet light of all wavelengths passes through hydrogen gas at room temperature, in the x-direction. Assume that all photons emitted due to electron transition inside the gas emerge in the y-direction. Let A and B denote the lights emerging from the gas in the x and y directions respectively.
(A) Some of the incident wavelengths will be absent in A.
(B) Only those wavelengths will be present in B which are absent in A.
(C) B will contain some visible light.
(D) B will contain some infrared light.

MP0107

41. In the hydrogen atom, if the reference level of potential energy is assumed to be zero at the ground state level. Choose the incorrect statement.
(A) The total energy of the shell increases with increase in the value of n
(B) The total energy of the shell decrease with increase in the value of n.
(C) The difference in total energy of any two shells remains the same.
(D) The total energy at the ground state becomes 13.6 eV.

MP0108

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42. A neutron collides head-on with a stationary hydrogen atom in ground state. Which of the following statements are correct (Assume that the hydrogen atom and neutron has same mass) :

(A) If kinetic energy of the neutron is less than 20.4 eV collision must be elastic.

(B) If kinetic energy of the neutron is less than 20.4 eV collision may be inelastic.

(C) Inelastic collision may be take place only when initial kinetic energy of neutron is greater than 20.4 eV.

(D) Perfectly inelastic collision can not take place.

MP0109

43. A free hydrogen atom in ground state is at rest. A neutron of kinetic energy 'K' collides with the hydrogen atom. After collision hydrogen atom emits two photons in succession one of which has energy 2.55 eV. (Assume that the hydrogen atom and neutron has same mass)

(A) minimum value of 'K' is 25.5 eV.

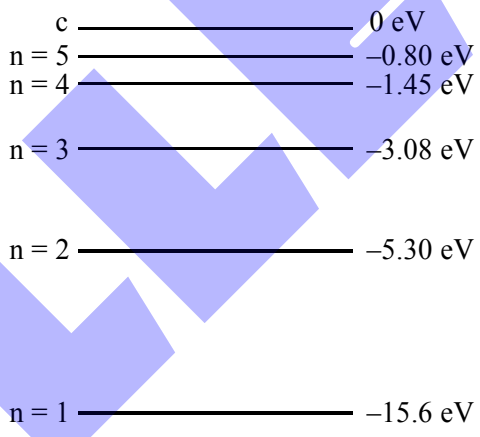
(B) minimum value of 'K' is 12.75 eV

(C) the other photon has energy 10.2 eV if K is minium.

(D) the upper energy level is of excitation energy 12.75 eV.

MP0110

44. The energy levels of a hypothetical one electron atom are shown in the figure



(A) The ionization potential of this atom is 15.6 V

(B) The short wavelength limit of the series terminating at n = 2 is 2339 Å

(C) The excitation potential for the state n = 3 is 12.52 V

(D) Wave number of the photon emitted for the transition n = 3 to n = 1 is 1.009×10^7 m^{-1}

MP0111

45. Suppose frequency of emitted photon is f_0 when electron of a stationary hydrogen atom jumps from a higher state m to a lower state n. If the atom is moving with a velocity v (<< c) and emits a photon of frequency f during the same transition, then which of the following statements are possible :-

(A) f may be equal to f_0 (B) f may be greater than f_0

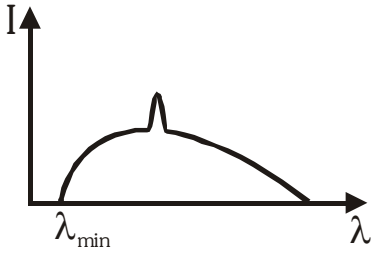
(C) f may be less than f_0 (D) f cannot be equal to f_0

MP0112

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D. X-Rays

46. An X-ray tube has three main controls.
(i) the target material (its atomic number Z)
(ii) the filament current (I_f)
(iii) the accelerating voltage (V)
Figure shows a typical intensity distribution against wavelength.



Which of the following is **CORRECT**?
(A) The limit λ_{min} is proportional to V^{-1}
(B) The sharp peak shifts to the right as Z is increased
(C) The penetrating power of X-ray increases if V is increased
(D) The intensity everywhere increases if filament current I_f is increased

MP0142

COMPREHENSION TYPE QUESTIONS

A. Photoelectric Effect

Paragraph for Question 47 to 49

A mercury arc lamp provides 0.1 watt of ultra-violet radiation at a wavelength of λ = 2537 Å only. The photo tube (cathode of photo electric device) consists of potassium and has an effective area of 4 cm^2. The cathode is located at a distance of 1m from the radiation source. The work function for potassium is ϕ_0 = 2.22 eV.

47. According to classical theory, the radiation from arc lamp spreads out uniformly in space as spherical wave. What time of exposure to the radiation should be required for a potassium atom (radius 2Å) in the cathode to accumulate sufficient energy to eject a photo-electron ?
(A) 352 second (B) 176 second (C) 704 seconds (D) No time lag

MP0143

48. To what saturation current does the flux of photons at the cathode corresponds if the photo conversion efficiency is 5%.
(A) 32.5 nA (B) 10.15 nA (C) 65 nA (D) 3.25 nA

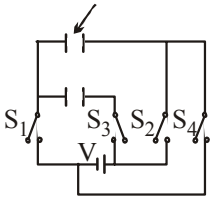
MP0143

49. What is the cut off potential V_0 ?
(A) 26.9 V (B) 2.69 V (C) 1.35 V (D) 5.33 V

MP0143

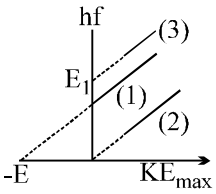
Paragraph for question nos. 50 to 52

The circuit shown is placed in vacuum. Both the capacitors are identical and they have the same capacitance C. Light is incident on the left plate of the upper capacitor. When all the switches are open then the hf versus KE_{max} is shown by the straight line (A). In all the cases, we are measuring the KE_{max} when the electron reaches the opposite plate.



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When only S_1 the switches and S_2 are closed, the graph becomes (B).
When only S_3 and S_4 are closed then the graph becomes (C).



50. What is the work function of the cathode?
(A) E (B) E_1 (C) $E + E_1$ (D) none of these
MP0144

51. What is the value of eV?
(A) E (B) E_1 (C) $E + E_1$ (D) none of these
MP0144

52. What is the value of E_1?
(A) 3E (B) 3E/2 (C) E/2 (D) none of these
MP0144

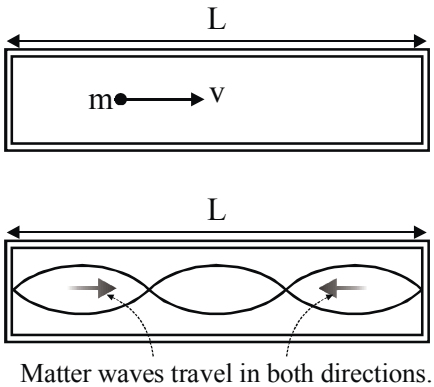
B. De-Broglie hypothesis

Paragraph for Question. 53 to 55

While conducting his doctoral research in theoretical physics and with no experimental evidence to go on, de Broglie reasoned by analogy with Einstein's equation E = hf and with some of the ideas of his theory of relativity. The details need not concern us, but they led de Broglie to postulate that if a material particle of momentum p = mv has a wave-like nature, then its wavelength must be given by

$\lambda = \frac{h}{p} = \frac{h}{mv}$ where h is Planck's constant. This is called the de-Broglie wavelength.

de-Broglie considered a matter wave to be a traveling wave. But suppose that a "particle" of matter is confined to a small region of space and cannot travel. How do the wave-like properties manifest themselves? This is the problem of "a particle in a box." Figure shows a particle of mass m moving in one dimension as it bounces back and forth with speed v between the ends of a box of length L. We'll call this a one-dimensional box; its width isn't relevant. A particle in a box creates a standing de Broglie wave as it reflects back and forth.



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53. What should be de-Broglie wavelength of confined particle in the box [here n∈N]

(A) $\frac{L}{2n}$ (B) $\frac{2L}{n}$ (C) $\frac{L}{n}$ (D) nL

MP0145

54. Confined particle's energy is given by

(A) $\frac{n^2h^2}{2mL^2}$ (B) $\frac{2n^2h^2}{mL^2}$ (C) $\frac{n^2h^2}{8mL^2}$ (D) $\frac{n^2h^2}{4mL^2}$

MP0145

55. Consider an oil drop from Millikan's oil drop experiment having diameter 1 μm confined between the plates separated by 10 μm. Density of oil is 900 kg/m^3. What is minimum energy of such an oil drop? [Given : h = 6.63 × 10^{-34} Js]

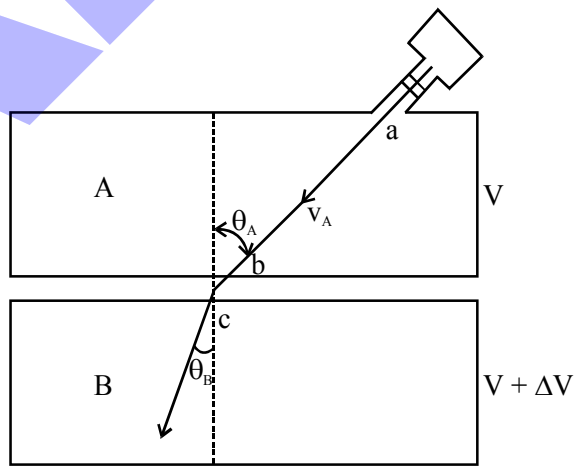
(A) 2.4 × 10^{-42} J (B) 1.2 × 10^{-42} J (C) 3.6 × 10^{-42} J (D) 4.8 × 10^{-42} J

MP0145

Paragraph for Question Nos. 56 to 58

Let a pencil of electrons from a suitable gun G enter through orifice in an enclosed metal box A, which has potential V relative to filament of the gun. Let these electrons emerge from A through orifice b and enter through c to another box B, which is maintained at a potential V + ΔV. The electric field between the two boxes changes the component of velocity of the electrons perpendicular to the adjacent surface, and the electrons enter B with a change in their direction of motion. Let v_A and v_B be the velocities of the electrons in A and B, respectively, and θ_A, θ_B the angles between these directions and the normal to the box faces at b and c. Since the electric field does not change the horizontal component of velocity,

$$v_A \sin\theta_A = v_B \sin\theta_B \qquad \frac{\sin\theta_A}{\sin\theta_B} = \frac{v_B}{v_A}$$



The "refraction" of a pencil of electrons

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Now if we dealing with light waves undergoing refraction, or any other kind of wave, the relation would be

$$\frac{\sin\theta_A}{\sin\theta_B} = \mu = \frac{u_A}{u_B}$$

where μ is the relative refractive index of the two media and u_A, u_B are the corresponding velocities of light wave. Comparison of the last two equations gives the result $\frac{u_A}{u_B} = \frac{v_B}{v_A}$. We may conclude that if matter waves follow the electron along its path, the wave speed u is inversely proportional to the speed v of the electron, or $u = \frac{b}{v}$.

56. If we define the frequency of matter waves as $f = \frac{u}{\lambda}$, the

(A) Frequency of matter waves in medium A is more than that in medium B
(B) Frequency of matter waves in medium A is less than that in medium B
(C) Frequency of matter waves in medium A is same as that in medium B
(D) Cannot be predicted

MP0146

57. Suppose $V_A = 20$ volt and $V_B = 15$ volt. Choose the **CORRECT** statement :-
(A) The speed of electrons as well as speed of matter waves inside box B is more
(B) The speed of electrons as well as speed of matter waves inside box B is less
(C) The speed of electrons inside the box B is more, but speed of matter waves in box B is less
(D) The speed of electrons inside the box B is less, but speed of matter waves in box B is more

MP0146

58. The refractive index for matter waves can be defined as $\frac{c}{u}$ where c is some constant. So refractive index for matter waves

(A) is inversely proportional to λ (B) is independent of λ
(C) is directly proportional of λ (D) is proportional to $\sqrt{\lambda}$

MP0146

C. Atomic structure

Paragraph for Question Nos. 59 to 61

The key feature of Bohr's theory of spectrum of hydrogen atom is the quantization of angular momentum when an electron is revolving around a proton. We will extend this to a general rotational motion to find quantized rotational energy of a diatomic molecule assuming it to be rigid. The rule to be applied is Bohr's quantization condition. **[JEE 2010]**

59. A diatomic molecule has moment of inertia I. By Bohr's quantization condition its rotational energy in the nth level (n = 0 is not allowed) is

(A) $\frac{1}{n^2}\left(\frac{h^2}{8\pi^2 I}\right)$ (B) $\frac{1}{n}\left(\frac{h^2}{8\pi^2 I}\right)$ (C) $n\left(\frac{h^2}{8\pi^2 I}\right)$ (D) $n^2\left(\frac{h^2}{8\pi^2 I}\right)$

MP0169

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60. It is found that the excitation frequency from ground to the first excited state of rotation for the CO molecule is close to $\frac{4}{\pi}\times10^{11}$ Hz. Then the moment of inertia of CO molecule about its center of mass is close to [Take $h = 2\pi \times 10^{-34}$ Js)

(A) 2.76×10^{-46} kg m^2 (B) 1.87×10^{-46} kg m^2

(C) 4.67×10^{-47} kg m^2 (D) 1.17×10^{-47} kg m^2

MP0169

61. In a CO molecule, the distance between C (mass = 12 a.m.u.) and O (mass = 16 a.m.u.), where 1 a.m.u. $= \frac{5}{3}\times10^{-27} kg$, is close to

(A) 2.4×10^{-10}m (B) 1.9×10^{-10}m

(C) 1.3×10^{-10}m (D) 4.4×10^{-11}m

MP0169

MATRIX MATCH TYPE QUESTION

A. Photoelectric Effect

62. Some quantities related to the photoelectric effect are mentioned under Column I and Column II. Match each quantity in Column I with the corresponding quantities in Column II on which it depends

Column-I	**Column-II**
(A) Saturation current	(P) Frequency of light
(B) Stopping potential	(Q) Work function
(C) de-Broglie wavelength of photoelectron	(R) Area of photosensitive plate
(D) Force due to radiation falling on the photo–plate.	(S) Intensity of light (at constant frequency)
	(T) None of these

MP0114

C. Atomic structure

63. When we write expression for energy of electron in n^{th} orbit of helium ion (He^+) we take zero potential energy for $n = \infty$, but the potential energy depends on reference. If we take total energy of atom for n = 1 orbit as zero then

Column–I	**Column–II**
(A) Total energy of electron in n = 2	(P) 54.4 eV
(B) Ionization energy from ground state	(Q) 40.8 eV
(C) Energy required to exit electron from n = 1 to n = 2	(R) depends on reference level
(D) Negative of potential energy of electron in n = 1	(S) independent of reference level.
	(T) 70.3 eV

MP0115

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64. In each situation of column–I a physical quantity related to orbiting electron in a hydrogen like atom is given. The terms 'Z' and 'n' given in column–II have usual meaning in Bohr's theory. Match the quantities in column–I with the terms they depend on in column–II :-

	Column-I		**Column-II**
(A)	Frequency of orbiting electron	(P)	Is directly proportional to Z^2
(B)	Angular momentum of orbiting electron	(Q)	Is directly proportional to n
(C)	Magnetic moment of orbiting electron	(R)	Is inversely proportional to n^3
(D)	The average current due to orbiting of electron	(S)	Is independent of Z
		(T)	None of these

MP0116

D. X-Rays

65. Match the entries of column–I with the entries of column–II :-

	Column-I		**Column-II**
(I)	Characteristic X-ray	(P)	Inverse process of photoelectric effect
(II)	Photoelectric effect	(Q)	Emission of electrons
(III)	Thermo-ionic emission	(R)	Moseley's law
(IV)	Continuous X-ray	(S)	Emission of radiations

Then choose the correct matching.

(A) (i) → (RS) ; (ii) → (RS) ; (iii) → (S) ; (iv) → (PS)

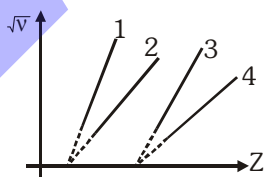
(B) (i) → (RS) ; (ii) → (Q) ; (iii) → (Q) ; (iv) → (PS)

(C) (i) → (RS) ; (ii) → (S) ; (iii) → (S) ; (iv) → (PRS)

(D) (i) → (RS) ; (ii) → (Q) ; (iii) → (Q) ; (iv) → (PRS)

MP0117

66. $\sqrt{\nu}$ versus Z graph for characteristic X–rays is as shown in figure. Match the following (assume screening constant for K_α and K_β is same and for L_α & L_β is same) :-



	Column-I		**Column-II**
(A)	Line – 1	(P)	L_α
(B)	Line – 2	(Q)	L_β
(C)	Line – 3	(R)	K_α
(D)	Line – 4	(S)	K_β
		(T)	Both K_α and L_β

MP0118

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EXERCISE (S)

A. Photoelectric Effect

1. In an experiment on photoelectric effect, the slope of the cut-off voltage versus frequency of incident light is found to be 4.12×10^{-15} V s. Calculate the value of Planck's constant. **NCERT**

MP0001

2. A 100W sodium lamp radiates energy uniformly in all directions. The lamp is located at the centre of a large sphere that absorbs all the sodium light which is incident on it. The wavelength of the sodium light is 589 nm. (a) What is the energy per photon associated with the sodium light ? (b) At what rate are the photons delivered to the sphere? **NCERT**

MP0002

3. In a photoelctric experiment set up, photons of energy 5 eV fall on the cathode having work function 3 eV.

(a) If the saturation current is $i_A = 4\mu A$ for intensity 10^{-5} W/m^2, then plot a graph between anode potential and current.

(b) Also draw a graph for intensity of incident radiation of 2×10^{-5} W/m^2 . **[JEE' 2003]**

MP0003

4. Monochromatic radiation of wavelength 640.2 nm (1nm = 10^{-9} m) from a neon lamp irradiates photosensitive material made of cesium. The stopping voltage is measured to be 0.54 V. The source is replaced by an iron source and its 427.2 nm line irradiates the same photo-cell. Predict the new stopping voltage. **NCERT**

MP0004

5. Monochromatic light of wavelength 632.8 nm is produced by a helium-neon laser. The power emitted is 9.42 mW. **NCERT**

(a) Find the energy and momentum of each photon in the light beam,

(b) How many photons per second, on the average, arrive at a target irradiated by this beam? (Assume the beam to have uniform cross-section which is less than the target area), and

(c) How fast does a hydrogen atom have to travel in order to have the same momentum as that of the photon ?

MP0005

6. A silver sphere of radius 1 cm and work function 4.7 eV is suspended from an insulating thread in free-space. It is under continuous illumination of 200 nm wavelength light. As photoelectrons are emitted, the sphere gets charged and acquires a potential. The maximum number of photoelectrons emitted from the sphere is $A \times 10^z$ (where $1 < A < 10$). The value of 'Z' is **[JEE 2011]**

MP0172

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B. De-Broglie hypothesis

7. Calculate the **NCERT**

(a) momentum, and

(b) de-Broglie wavelength of the electrons accelerated through a potential difference of 56 V.

MP0006

8. The wavelength of light from the spectral emission line of sodium is 589 nm. Find the kinetic energy at which **NCERT**

(a) an electron, and

(b) a neutron, would have the same de-Broglie wavelength.

MP0007

9. (a) For what kinetic energy of a neutron will the associated de-Broglie wavelength be 1.40×10^{-10} m?

(b) Also find the de-Broglie wavelength of a neutron, in thermal equilibrium with matter, having an average kinetic energy of (3/2) k T at 300 K. **NCERT**

MP0008

10. The potential energy of a particle varies as $U(x) = E_0$ for $0 \le x \le 1$ and $U(x) = 0$ for $x > 1$

For $0 \le x \le 1$. de–Broglie wavelengths is λ_1 and for $x > 1$ the de–Broglie wavelength is λ_2.

Total energy of the particle is $2E_0$. Find $\frac{\lambda_1}{\lambda_2}$. **[JEE 2005]**

MP0009

11. Atomic furnace wall has a hole having a diameter of δ. At a distance of L from the furnace wall, we kept a plane screen which can detect atoms striking it. Assume that atoms of mass m moving with velocity v in x direction emerge from the hole. Using de Broglie's hypothesis, find the smallest radius (in mm) of the spot on the screen. (Take : $h = 6.4 \times 10^{-34}$ kg m^2/s, L = 1m, $m = 1.6 \times 10^{-27}$ kg, $v = 10^3$ m/s, $\delta = 0.2$ µm.

MP0386

12. A proton is fired from very far away towards a nucleus with charge Q = 120 e, where e is the electronic charge. It makes a closest approach of 10 fm to the nucleus. The de Broglie wavelength (in units of fm) of the proton at its start is (Take : The proton mass, $m_p = (5/3) \times 10^{-27}$ kg; $h/e = 4.2 \times 10^{-15}$ J.s/C; $\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9$ m/F; 1 fm = 10^{-15} m) **[JEE 2012]**

MP0173

13. A particle of mass m is confined to a narrow tube of length L.

(a) Find the wavelengths of the de–Broglie wave which will resonate in the tube.

(b) Calculate the corresponding particle momenta, and

(c) Calculate the corresponding energies.

MP0403

C. Atomic structure

14. Determine the number of lines in Paschen series which have a wavelength greater than 1000 nm.

MP0010

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15. In a photoelectric setup, the radiations from the Balmer series of hydrogen atom are incident on a metal surface of work function 2eV. The wavelength of incident radiations lies between 450 nm to 700 nm. Find the maximum kinetic energy of photoelectron emitted. (Given hc/e = 1242 eV-nm).

[JEE-2004]

MP0011

16. An electron and a proton are separated by a large distance and the electron approaches the proton with a kinetic energy of 4.11 eV. If the electron is captured by the proton to form hydrogen atom in the ground state, the wavelength of photon given off is $\alpha \times 10^2$ Å? Fill the value of α in your OMR sheet.

MP0012

17. A neutron moving through container filled with stationary deuterons. The neutron successively collides elastically and head on with stationary deuterons one at a time. The mass of the neutron is equal to half that of the deuteron. How many such collision would be required to slow the neutron down from 81 eV to 1 eV. [Neglect the relativistic effect]

MP0013

18. What is the shortest wavelength present in the Paschen series of spectral lines? **NCERT**

MP0014

19. A 12.5 eV electron beam is used to bombard gaseous hydrogen at room temperature. What series of wavelengths will be emitted? **NCERT**

MP0015

20. The total energy of an electron in the first excited state of the hydrogen atom is about –3.4 eV.

(a) What is the kinetic energy of the electron in this state? **NCERT**

(b) What is the potential energy of the electron in this state?

(c) Which of the answers above would change if the choice of the zero of potential energy is changed?

MP0016

21. Obtain the first Bohr's radius and the ground state energy of a muonic hydrogen atom [i.e., an atom in which a negatively charged muon (μ^-) of mass about $207m_e$ orbits around a proton]. **NCERT**

MP0017

22. A potential difference of V volts is applied on two parallel electrodes separated by a distance of 4.0×10^{-2} m. The electrons of very low energy are injected into the region between the electrodes which contains argon at low pressure . The average distance the electrons travel between collisions with argon atoms is 8×10^{-5} m . The ionization energy of argon atom is 16 eV . Estimate the minimum value of V(in kV) such that the electrons will cause ionization in argon atoms by collision .

MP0018

23. In hydrogen-like atom (Z = 11), n^{th} line of Lyman series has wavelength λ equal to the de-Broglie's wavelength of electron in the level from which it originated. What is the value of n?

[Take: Bohr radius (r_0) = 0.53 Å and Rydberg constant (R) = 1.1×10^7 m^{-1}] **[JEE 2006]**

MP0019

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D. X-Rays

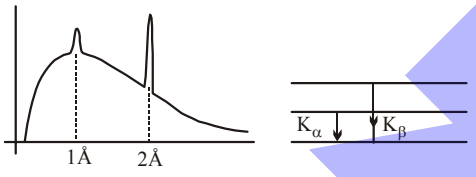
24. The wavelength of characteristic K_α-line emitted by a hydrogen like element is 2.5 Å. Find the wavelength of the K_γ-line emitted by the same element (in Å). [Assume the shielding effect to be same as of K_α]

MP0020

25. In an accelerator experiment on high-energy collisions of electrons with positrons, a certain event is interpreted as annihilation of an electron-positron pair of total energy 10.2 BeV into two γ-rays of equal energy. What is the wavelength associated with each γ-ray? (1BeV = 10^9 eV) **NCERT**

MP0021

26. Figure shows K_α & K_β X-rays along with continuous X-ray. Find the energy of L_α X-ray. (Use hc = 12420 evÅ).



MP0022

27. A graph of $\sqrt{\nu}$ (where ν is the frequency of K_α line of the characteristic X-ray spectrum) is plotted against the atomic number Z of the elements emitting the characteristic X-ray. The intercept of the graph on the Z-axis is 1 and the slope of the graph is 0.5×10^8 S.I. units. The frequency of the K_α line for an element of atomic number 41 is given as $\alpha \times 10^{16}$ Hz. Find the value of α.

MP0023

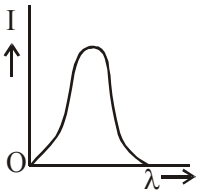
28. A nickel crystal is used as a diffraction grating for x-rays. Then the same crystal is used to diffract electrons. If the two diffraction patterns are identical, and the energy of each x-ray photon is E = 20.0keY. What is the kinetic energy of each electron?

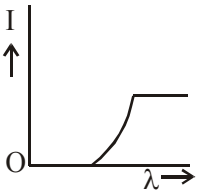
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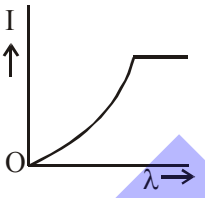
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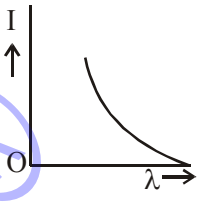
EXERCISE (JM)

1. The anode voltage of photocell is kept fixed. The wavelength λ of the light falling on the cathode is gradually changed. The plate current I of the photocell varies as follows : **[AIEEE-2013]**

(1) 

(2) 

(3) 

(4) 

MP0155

2. In a hydrogen like atom electron makes transition from an energy level with quantum number n to another with quantum number (n–1). If n >> 1, the frequency of radiation emitted is proportional to : **[JEE Main-2013]**

(1) $\frac{1}{n}$ (2) $\frac{1}{n^2}$ (3) $\frac{1}{n^{3/2}}$ (4) $\frac{1}{n^3}$

MP0156

3. Hydrogen $({}_1H^1)$, Deuterium $({}_1H^2)$, singly ionised Helium $({}_2He^4)^+$ and doubly ionised lithium $({}_3Li^6)^{++}$ all have one electron around the nucleus. Consider an electron transition from n = 2 to n = 1. If the wave lengths of emitted radiation are λ_1, λ_2, λ_3 and λ_4 respectively then approximately which one of the following is **correct** ? **[JEE-Main-2014]**

(1) $\lambda_1 = \lambda_2 = 4\lambda_3 = 9\lambda_4$ (2) $\lambda_1 = 2\lambda_2 = 3\lambda_3 = 4\lambda_4$

(3) $4\lambda_1 = 2\lambda_2 = 2\lambda_3 = \lambda_4$ (4) $\lambda_1 = 2\lambda_2 = 2\lambda_3 = \lambda_4$

MP0285

4. The radiation corresponding to 3 → 2 transition of hydrogen atom falls on a metal surface to produce photoelectrons. These electrons are made to enter a magnetic field of 3×10^{-4}T. If the radius of the largest circular path followed by these electrons is 10.0 mm, the work function of the metal is close to:- **[JEE-Main-2014]**

(1) 0.8 eV (2) 1.6 eV (3) 1.8 eV (4) 1.1 eV

MP0286

5. As an electron makes a transition from an excited state to the ground state of a hydrogen - like atom/ion : **[JEE Main-2015]**

(1) kinetic energy decreases, potential energy increases but total energy remains same

(2) kinetic energy and total energy decrease but potential energy increases

(3) its kinetic energy increases but potential energy and total energy decreases

(4) kinetic energy, potential energy and total energy decrease

MP0157

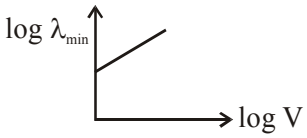
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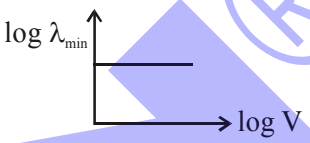
6. Radiation of wavelength λ, is incident on a photocell. The fastest emitted electron has speed *v*. If the wavelength of changed to $\frac{3\lambda}{4}$, the speed of the fastest emitted electron will be :-**[JEE Main-2016]**

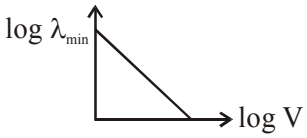
(1) $= v\left(\frac{3}{5}\right)^{1/2}$ (2) $> v\left(\frac{4}{3}\right)^{1/2}$ (3) $< v\left(\frac{4}{3}\right)^{1/2}$ (4) $= v\left(\frac{4}{3}\right)^{1/2}$

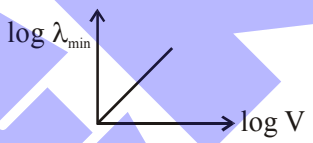
MP0159

7. An electron beam is accelerated by a potential difference V to hit a metallic target to produce X-rays. It produces continuous as well as characteristic X-rays. If λ_{min} is the smallest possible wavelength of X-ray in the spectrum, the variation of log λ_{min} with log V is correctly represented in : **[JEE Main-2017]**

(1) 

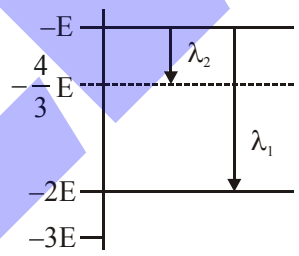
(2) 

(3) 

(4) 

MP0160

8. Some energy levels of a molecule are shown in the figure. The ratio of the wavelengths $r = \lambda_1/\lambda_2$, is given by : **[JEE Main-2017]**



(1) $r = \frac{3}{4}$ (2) $r = \frac{1}{3}$ (3) $r = \frac{4}{3}$ (4) $r = \frac{2}{3}$

MP0161

9. A particle A of mass m and initial velocity v collides with a particle B of mass $\frac{m}{2}$ which is at rest. The collision is head on, and elastic. The ratio of the de–Broglie wavelengths λ_A to λ_B after the collision is: **[JEE Main-2017]**

(1) $\frac{\lambda_A}{\lambda_B} = \frac{2}{3}$ (2) $\frac{\lambda_A}{\lambda_B} = \frac{1}{2}$ (3) $\frac{\lambda_A}{\lambda_B} = \frac{1}{3}$ (4) $\frac{\lambda_A}{\lambda_B} = 2$

MP0162

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10. An electron from various excited states of hydrogen atom emit radiation to come to the ground state. Let λ_n, λ_g be the de Broglie wavelength of the electron in the nth state and the ground state respectively. Let Λ_n be the wavelength of the emitted photon in the transition from the nth state to the ground state. For large n, (A, B are constants) **[JEE-Main-2018]**

(1) $\Lambda_n \approx A + B\lambda_n$ (2) $\Lambda_n^2 \approx A + B\lambda_n^2$ (3) $\Lambda_n^2 \approx \lambda$ (4) $\Lambda_n \approx A + \frac{B}{\lambda_n^2}$

MP0289

11. If the series limit frequency of the Lyman series is v_L, then the series limit frequency of the Pfund series is : **[JEE Main-2018]**

(1) 16 v_L (2) $v_L/16$ (3) $v_L/25$ (4) 25 v_L

MP0163

SELECTED PROBLEMS FROM JEE-MAINS ONLINE PAPERS

12. A hydrogen atom, initially in the ground state is excited by absorbing a photon of wavelength 980Å. The radius of the atom in the excited state, it terms of Bohr radius a_0, will be :
(h_c = 12500 eV – Å) **[JEE Main-2019_Jan]**

(1) $9a_0$ (2) $25a_0$ (3) $4a_0$ (4) $16a_0$

MP0356

13. When a certain photosensistive surface is illuminated with monochromatic light of frequency ν, the stopping potential for the photo current is $-V_0/2$. When the surface is illuminated by monochromatic light of frequency ν/2, the stopping potential is $-V_0$. The threshold frequency for photoelectric emission is: **[JEE Main-2019_Jan]**

(1) $\frac{3\nu}{2}$ (2) 2ν (3) $\frac{4}{3}\nu$ (4) $\frac{5\nu}{3}$

MP0357

14. An alpha-particle of mass m suffers 1-dimensional elastic coolision with a nucleus at rest of unknown mass. It is scattered directly backwards losing, 64% of its initial kinetic energy. The mass of the nucleus is :- **[JEE Main-2019_Jan]**

(1) 4 m (2) 3.5 m (3) 2 m (4) 1.5 m

MP0358

15. In a Frank-Hertz experiment, an electron of energy 5.6 eV passes through mercury vapour and emerges with an energy 0.7 eV. The minimum wavelength of photons emitted by mercury atoms is close to :- **[JEE Main-2019_Jan]**

(1) 2020 nm (2) 220 nm (3) 250 nm (4) 1700 nm

MP0359

16. A nucleus A, with a finite de-broglie wavelength λ_A, undergoes spontaneous fission into two nuclei B and C of equal mass. B flies in the same direction as that of A, while C flies in the opposite direction with a velocity equal to half of that of B. The de-Broglie wavelengths λ_B and λ_C of B and C are respectively :- **[JEE Main-2019_April]**

(1) $2\lambda_A, \lambda_A$ (2) $\lambda_A, 2\lambda_A$ 3) $\lambda_A, \frac{\lambda_A}{2}$ (4) $\frac{\lambda_A}{2}, \lambda_A$

MP0360

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17. Two particles move at right angle to each other. Their de-Broglie wavelengths are λ_1 and λ_2 respectively. The particles suffer perfectly inelastic collision. The de-Broglie wavelength λ, of the final particle, is given by : **[JEE Main-2019_April]**

(1) $\lambda = \frac{\lambda_1 + \lambda_2}{2}$ (2) $\frac{2}{\lambda} = \frac{1}{\lambda_1} + \frac{1}{\lambda_2}$ (3) $\lambda = \sqrt{\lambda_1 \lambda_2}$ (4) $\frac{1}{\lambda^2} = \frac{1}{\lambda_1^2} + \frac{1}{\lambda_2^2}$

MP0361

18. 50 W/m^2 energy density of sunlight is normally incident on the surface of a solar panel. Some part of incident energy (25%) is reflected from the surface and the rest is absorbed. The force exerted on 1m^2 surface area will be close to ($c = 3 \times 10^8$ m/s) :- **[JEE Main-2019_April]**

(1) 15×10^{-8} N (2) 35×10^{-8} N (3) 10×10^{-8} N (4) 20×10^{-8} N

MP0362

19. Light is incident normally on a completely absorbing surface with an energy flux of 25 Wcm^{-2}. if the surface has an area of 25 cm^{-2}, the momentum transferred to the surface in 40 min time duration will be : **[JEE Main-2019_April]**

(1) 5.0×10^{-3} Ns (2) 3.5×10^{-6} Ns (3) 1.4×10^{-6} Ns (4) 6.3×10^{-4} Ns

MP0363

20. In Li^{++}, electron in first Bohr orbit is excited to a level by a radiation of wavelength λ. when the ion gets deexcited to the ground state in all possible ways (including intermediate emissions), a total of six spectral lines are observed. What is the value of λ ? (Given : $h = 6.63 \times 10^{-34}$ Js ; $c = 3 \times 10^8$ ms^{-1}) **[JEE Main-2019_April]**

(1) 9.4 nm (2) 12.3 nm (3) 10.8 nm (4) 11.4 nm

MP0364

21. In a photoelectric effect experiment the threshold wavelength of the light is 380 nm. If the wavelentgh of incident light is 260 nm, the maximum kinetic energy of emitted electrons will be: Given E (in eV) $= \frac{1237}{\lambda(\text{in nm})}$ **[JEE Main-2019_April]**

(1) 1.5 eV (2) 4.5 eV (3) 15.1 eV (4) 3.0 eV

MP0365

22. Consider an electron in a hydrogen atom, revolving in its second excited state (having radius 4.65Å). The de-Broglie wavelength of this electron is : **[JEE Main-2019_April]**

(1) 12.9 Å (2) 3.5 Å (3) 9.7 Å (4) 6.6 Å

MP0366

23. An excited He^+ ion emits two photons in succession, with wavelengths 108.5 nm and 30.4 nm, in making a transition to ground state. The quantum number n, corresponding to its initial excited state is (for photon of wavelength λ, energy $E = \frac{1240\,\text{eV}}{\lambda(\text{in nm})}$) : **[JEE Main-2019_April]**

(1) n = 5 (2) n = 4 (3) n = 6 (4) n = 7

MP0367

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24. A beam of electromagnetic radiation of intensity 6.4×10^{-5} W/cm^2 is comprised of wavelength, $\lambda = 310$ nm. It falls normally on a metal (work function $\varphi = 2$eV) of surface area of 1 cm^2. If one in 10^3 photons ejects an electron, total number of electrons ejected in 1 s is 10^x. (hc = 1240 eVnm, 1eV = 1.6×10^{-19} J), then x is____. **[JEE Main-2020_Jan]**

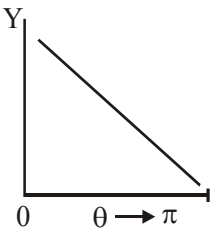
MP0368

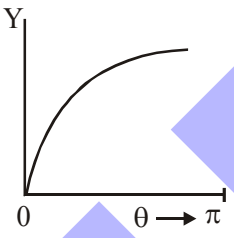
25. An electron (of mass m) and a photon have the same energy E in the range of a few eV. The ratio of the de-Broglie wavelength associated with the electron and the wavelength of the photon is (c = speed of light in vacuum) **[JEE Main-2020_Jan]**

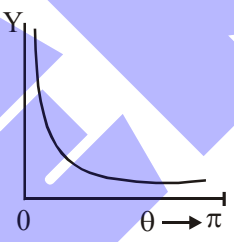
(1) $\left(\frac{E}{2m}\right)^{1/2}$ (2) $\frac{1}{c}\left(\frac{E}{2m}\right)^{1/2}$ (3) $c(2mE)^{1/2}$ (4) $\frac{1}{c}\left(\frac{2E}{m}\right)^{1/2}$

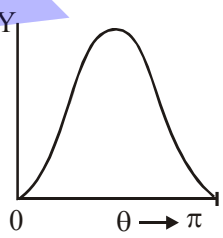
MP0369

26. The graph which depicts the results of Rutherform gold foil experiment with α-particales is :
θ : Scattering angle
Y : Number of scattered α-particles detected (Plots are schematic and not to scale)
[JEE Main-2020_Jan]

(1) 

(2) 

(3) 

(4) 

MP0370

27. An electron (mass m) with initial velocity $\vec{v} = v_0\hat{i} + v_0\hat{j}$ is in an electric field $\vec{E} = -E_0\hat{k}$. If λ_0 is initial de-Broglie wavelength of electron, its de-Broglie wave length at time t is given by : **[JEE Main-2020_Jan]**

(1) $\frac{\lambda_0\sqrt{2}}{\sqrt{1+\frac{e^2E^2t^2}{m^2v_0^2}}}$ (2) $\frac{\lambda_0}{\sqrt{2+\frac{e^2E^2t^2}{m^2v_0^2}}}$ (3) $\frac{\lambda_0}{\sqrt{1+\frac{e^2E^2t^2}{2m^2v_0^2}}}$ (4) $\frac{\lambda_0}{\sqrt{1+\frac{e^2E_0^2t^2}{m^2v_0^2}}}$

MP0371

28. The energy required to ionise a hydrogen like ion in its ground state is 9 Rydbergs. What is the wavelength of the radiation emitted when the electron in this ion jumps from the second excited state to the ground state ? **[JEE Main-2020_Jan]**

(1) 35.8 nm (2) 24.2 nm (3) 8.6 nm (4) 11.4 nm

MP0372

29. In a hydrogen atom the electron makes a transition from $(n + 1)^{th}$ level to the nth level. If n>>l, the frequency of radiation emitted is proportional to : **[JEE Main-2020_Jan]**

(1) $\frac{1}{n^4}$ (2) $\frac{1}{n^3}$ (3) $\frac{1}{n^2}$ (4) $\frac{1}{n}$

MP0373

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30. Particle A of mass $m_A = \frac{m}{2}$ moving along the x-axis with velocity v_0 collides elastically with another particle B at rest having mass $m_B = \frac{m}{3}$. If both particles move along the x-axis after the collision, the change $\Delta\lambda$ in de-Broglie wavelength of particle A, in terms of its de-Broglie wavelength (λ_0) before collision is : **[JEE Main-2020_Sep]**

(1) $\Delta\lambda = 4\lambda_0$ (2) $\Delta\lambda = \frac{5}{2}\lambda_0$ (3) $\Delta\lambda = 2\lambda_0$ (4) $\Delta\lambda = \frac{3}{2}\lambda_0$

MP0374

31. In the line spectra of hydrogen atom, difference between the largest and the shortest wavelengths of the Lyman series is 304 Å. The corresponding difference for the Paschan series in Å is : _____. **[JEE Main-2020_Sep]**

MP0375

32. A particle of mass 200 MeV/c^2 collides with a hydrogen atom at rest. Soon after the collision the particle comes to rest, and the atom recoils and goes to its first excited state. The initial kinetic energy of the particle (in eV) is $\frac{N}{4}$. The value of N is : **[JEE Main-2020_Sep]**

(Given the mass of the hydrogen atom to be 1 GeV/c^2) _____.

MP0376

33. Assuming the nitrogen molecule is moving with r.m.s. velocity at 400 K, the de-Broglie wavelength of nitrogen molecule is close to :
(Given : nitrogen molecule weight : 4.64×10^{-26}kg, Boltzman constant : 1.38×10^{-23} J/K, Planck constant: 6.63×10^{-34} J.s) **[JEE Main-2020_Sep]**

(1) 0.34 Å (2) 0.24 Å (3) 0.20 Å (4) 0.44 Å

MP0377

34. A beam of electrons of energy E scatters from a target having atomic spacing of 1Å. The first maximum intensity occurs at $\theta = 60°$. Then E (in eV) is _____.
(Planck constant $h = 6.64 \times 10^{-34}$ Js, $1eV = 1.6 \times 10^{-19}$ J, electron mass $m = 9.1 \times 10^{-31}$ kg)

[JEE Main-2020_Sep]

WO0163

35. Given below are two statements :
Statement-I : Two photons having equal linear momenta have equal wavelengths.
Statement-II : If the wavelength of photon is decreased, then the momentum and energy of a photon will also decrease. **[JEE Main-2021_Feb]**
In the light of the above statements, choose the correct answer from the options given below.
(1) Both Statement I and Statement II are true
(2) Statement I is false but Statement II is true
(3) Both Statement I and Statement II are false
(4) Statement I is true but Statement II is false

MP0387

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36. Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R.
Assertion A : An electron microscope can achieve better resolving power than an optical microscope.
Reason R : The de Broglie's wavelength of the electrons emitted from an electron gun is much less than wavelength of visible light. **[JEE Main-2021_Feb]**
In the light of the above statements, choose the correct answer from the options given below:
(1) A is true but R is false.
(2) Both A and R are true and R is the correct explanation of A.
(3) Both A and R are true but R is NOT the correct explanation of A.
(4) A is false but R is true.

MP0388

37. The recoil speed of a hydrogen atom after it emits a photon in going from n = 5 state to n = 1 state will be : **[JEE Main-2021_Feb]**
(1) 4.17 m/s (2) 2.19 m/s (3) 3.25 m/s (4) 4.34 m/s

MP0389

38. If 2.5×10^{-6} N average force is exerted by a light wave on a non-reflecting surface of 30 cm^2 area during 40 minutes of time span, the energy flux of light just before it falls on the surface is ____ W/cm^2. (Round off to the Nearest Integer)
(Assume complete absorption and normal incidence conditions are there) **[JEE Main-2021_March]**

MP0390

39. A particle of mass m moves in a circular orbit in a central potential field $U(r) = U_0 r^4$. If Bohr's quantization conditions are applied, radii of possible orbitals r_n vary with $n^{1/\alpha}$, where α is_____.
[JEE Main-2021_March]

MP0391

40. The speed of electrons in a scanning electron microscope is 1×10^7 ms^{-1}. If the protons having the same speed are used instead of electrons, then the resolving power of scanning proton microscope will be changed by a factor of: **[JEE Main-2021_March]**
(1) 1837 (2) $\frac{1}{1837}$ (3) $\sqrt{1837}$ (4) $\frac{1}{\sqrt{1837}}$

MP0392

41. The radiation corresponding to $3 \rightarrow 2$ transition of a hydrogen atom falls on a gold surface to generate photoelectrons. These electrons are passed through a magnetic field of 5×10^{-4} T. Assume that the radius of the largest circular path followed by these electrons is 7 mm, the work function of the metal is : (Mass of electron = 9.1×10^{-31} kg) **[JEE Main-2021_July]**
(1) 1.36 eV (2) 1.88 eV (3) 0.16 eV (4) 0.82 eV

MP0393

42. Consider two separate ideal gases of electrons and protons having same number of particles. The temperature of both the gases are same. The ratio of the uncertainty in determining the position of an electron to that of a proton is proportional to :- **[JEE Main-2021_Aug]**
(1) $\left(\frac{m_p}{m_e}\right)^{3/2}$ (2) $\sqrt{\frac{m_e}{m_p}}$ (3) $\sqrt{\frac{m_p}{m_e}}$ (4) $\frac{m_p}{m_e}$

MP0394

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43. In Bohr's atomic model of hydrogen, let K. P and E are the kinetic energy, potential energy and total energy of the electron respectively. Choose the correct option when the electron undergoes transitions to a higher level : **[JEE Main-2022_June]**

(A) All K. P and E increase.
(B) K decreases P and E increase.
(C) P decreases K and E increase.
(D) K increases P and E decrease. **MP0404**

44. Choose the correct option from the following options given below : **[JEE Main-2022_June]**

(A) In the ground state of Rutherford's model electrons are in stable equilibrium. While in Thomson's model electrons always experience a net-force.
(B) An atom has a nearly continuous mass distribution in a Rutherford's model but has a highly non-uniform mass distribution in Thomson's model
(C) A classical atom based on Rutherford's model is doomed to collapse.
(D) The positively charged part of the atom possesses most of the mass in Rutherford's model but not in Thomson's model. **MP0405**

45. A proton, a neutron, an electron and an α-particle have same energy. If λ_p, λ_n, λ_e and λ_α are the de Broglie's wavelengths of proton, neutron, electron and α-particle respectively, then choose the correct relation from the following : **[JEE Main-2022_June]**

(A) $\lambda_p = \lambda_n > \lambda_e > \lambda_\alpha$
(B) $\lambda_\alpha < \lambda_n < \lambda_p < \lambda_e$
(C) $\lambda_e < \lambda_p = \lambda_n > \lambda_\alpha$
(D) $\lambda_e = \lambda_p = \lambda_n = \lambda_\alpha$ **MP0406**

46. The ratio for the speed of the electron in the 3rd orbit of He^+ to the speed of the electron in the 3rd orbit of hydrogen atom will be :- **[JEE Main-2022_June]**

(A) 1 : 1 (B) 1 : 2 (C) 4 : 1 (D) 2 : 1 **MP0407**

47. A metal surface is illuminated by a radiation of wavelength 4500 Å. The ejected photo-electron enters a constant magnetic field of 2 mT making an angle of 90° with the magnetic field. If it starts revolving in a circular path of radius 2 mm, the work function of the metal is approximately : **[JEE Main-2022_June]**

(A) 1.36 eV (B) 1.69 eV (C) 2.78 eV (D) 2.23 eV **MP0408**

48. A hydrogen atom in is ground state absorbs 10.2 eV of energy. The angular momentum of electron of the hydrogen atom will increase by the value of : (Given, Plank's constant = 6.6×10^{-34} Js) **[JEE Main-2022_June]**

(A) $2.10 \times 10^{-34}\ Js$ (B) $1.05 \times 10^{-34}\ Js$ (C) $3.15 \times 10^{-34}\ Js$ (D) $4.2 \times 10^{-34}\ Js$ **MP0409**

49. A beam of monochromatic light is used to excite the electron in Li^{++} from the first orbit to the third orbit. The wavelength of monochromatic light is found to be $x \times 10^{-10}$m. The value of x is _____ . [Given hc = 1242 eV nm] **[JEE Main-2022_June]** **MP0410**

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50. Given below are two statements : one is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A : The photoelectric effect does not take place, if the energy of the incident radiation is less than the work function of a metal.

Reason R : Kinetic energy of the photoelectrons is zero, if the energy of the incident radiation is equal to the work function of a metal.

In the light of the above statements, choose the most appropriate answer from the options given below.

[JEE Main-2022_June]

(A) Both **A** and **R** are correct and **R** is the correct explanation of **A**

(B) Both **A** and **R** are correct but **R** is not the correct explanation of **A**

(C) **A** is correct but **R** is not correct

(D) **A** is not correct but **R** is correct

MP0411

51. $\sqrt{d_1}$ and $\sqrt{d_2}$ are the impact parameters corresponding to scattering angles 60° and 90° respectively, when an α particle is approaching a gold nucleus. For $d_1 = x\, d_2$, the value of x will be_____.

[JEE Main-2022_June]

MP0412

52. A metal exposed to light of wavelength 800 nm and emits photoelectrons with a certain kinetic energy. The maximum kinetic energy of photo-electron doubles when light of wavelength 500 nm is used. The work function of the metal is (Take hc = 1230 eV-nm). **[JEE Main-2022_July]**

(A) 1.537 eV (B) 2.46 eV (C) 0.615 eV (D) 1.23 eV

MP0413

53. The magnetic moment of an electron (e) revolving in an orbit around nucleus with an orbital angular momentum is given by : **[JEE Main-2022_July]**

(1) $\vec{\mu}_L = \frac{e\vec{L}}{2m}$ (2) $\vec{\mu}_L = -\frac{e\vec{L}}{2m}$ (3) $\vec{\mu}_l = -\frac{e\vec{L}}{m}$ (4) $\vec{\mu}_l = \frac{2e\vec{L}}{m}$

MP0414

54. Hydrogen atom from excited state comes to the ground by emitting a photon of wavelength λ. The value of principal quantum number 'n' of the excited state will be : **[JEE Main-2022_July]**

(R : Rydberg constant)

(A) $\sqrt{\frac{\lambda R}{\lambda - 1}}$ (B) $\sqrt{\frac{\lambda R}{\lambda R - 1}}$ (C) $\sqrt{\frac{\lambda}{\lambda R - 1}}$ (D) $\sqrt{\frac{\lambda R^2}{\lambda R - 1}}$

MP0415

55. A parallel beam of light of wavelength 900 nm and intensity 100 Wm^{-2} is incident on a surface perpendicular to the beam. Tire number of photons crossing 1 cm^2 area perpendicular to the beam in one second is : **[JEE Main-2022_July]**

(A) 3×10^{16} (B) 4.5×10^{16} (C) 4.5×10^{17} (D) 4.5×10^{20}

MP0416

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56. An electron (mass m) with an initial velocity $\vec{v} = v_0\hat{i}(v_0 > 0)$ is moving in an electric field $\vec{E} = -E_0\hat{i}(E_0 > 0)$ where E_0 is constant. If at t = 0 de Broglie wavelength is $\lambda_0 = \frac{h}{mv_0}$, then its de Broglie wavelength after time t is given by **[JEE Main-2022_July]**

(A) λ_0 (B) $\lambda_0\left(1+\frac{eE_0t}{mv_0}\right)$ (C) $\lambda_0 t$ (D) $\frac{\lambda_0}{\left(1+\frac{eE_0t}{mv_0}\right)}$

MP0417

57. The equation $\lambda = \frac{1.227}{x}$ nm can be used to find the de-Broglie wavelength of an electron. In this equation x stands for : **[JEE Main-2022_July]**

Where,
m = mass of electron
P = momentum of electron
K = Kinetic energy of electron
V = Accelerating potential in volts for electron

(A) $\sqrt{mK}$ (B) $\sqrt{P}$ (C) $\sqrt{K}$ (D) $\sqrt{V}$

MP0418

58. Nearly 10% of the power of a 110 W light bulb is converted to visible radiation. The change in average intensities of visible radiation, at a distance of 1 m from the bulb to a distance of 5 m is $a \times 10^{-2}$ W/m^2. The value of 'a' will be **[JEE Main-2022_July]**

MP0419

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EXERCISE (JA)

1. The work functions of Silver and sodium are 4.6 and 2.3 eV, repetitively. The ratio of the slope of the stopping potential versus frequency plot for Silver to that of Sodium is. **[JEE Advanced-2013]**

MP0174

2. The radius of the orbit of an electron in a Hydrogen-like atom is 4.5 a_0, where a_0 is the Bohr radius. Its orbital angular momentum is $\frac{3h}{2\pi}$. It is given that h is Planck constant and R is Rydberg constant. The possible wavelength (s), when the atom de-excites, is (are) :- **[JEE Advanced-2013]**

(A) $\frac{9}{32R}$ (B) $\frac{9}{16R}$ (C) $\frac{9}{5R}$ (D) $\frac{4}{3R}$

MP0175

3. If λ_{Cu} is the wavelength of K_α X-ray line of copper (atomic number 29) and λ_{Mo} is the wavelength of the K_α X-ray line of molybdenum (atomic number 42), then the ratio $\frac{\lambda_{Cu}}{\lambda_{Mo}}$ is close to :-

[JEE Advance-2014]

(A) 1.99 (B) 2.14 (C) 0.50 (D) 0.48

MP0299

4. A metal surface is illuminated by light of two different wavelength 248 nm and 310 nm. The maximum speeds of the photoelectrons corresponding to these wavelengths are u_1 and u_2, respectively. If the ratio $u_1 : u_2 = 2 : 1$ and hc = 1240 eV nm, the work function of the metal is nearly :-

[JEE Advance-2014]

(A) 3.7 eV (B) 3.2 eV (C) 2.8 eV (D) 2.5 eV

MP0300

5. Consider a hydrogen atom with its electron in the nth orbital. An electromagnetic radiation of wavelength 90 nm is used to ionize the atom. If the kinetic energy of the ejected electron is 10.4 eV, then the value of n is (hc = 1242 eV nm). **[JEE Advanced-2015]**

MP0176

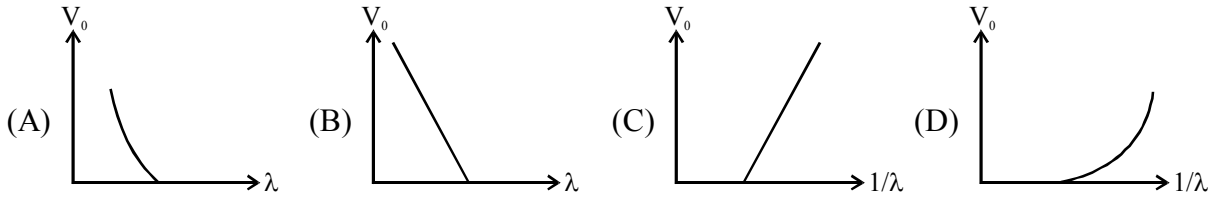
6. Planck's constant h, speed of light c and gravitational constant G are used to form a unit of length L and a unit of mass M. Then the correct option(s) is(are) :- **[JEE Advanced-2015]**

(A) $M \propto \sqrt{c}$ (B) $M \propto \sqrt{G}$ (C) $L \propto \sqrt{h}$ (D) $L \propto \sqrt{G}$

MP0177

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7. For photo-electric effect with incident photon wavelength λ, the stopping potential is V_0. Identify the correct variation(s) of V_0 with λ and 1/λ. **[JEE Advanced-2015]**



MP0178

8. An electron in an excited state of Li^{2+} ion has angular momentum $3h/2\pi$. The de Broglie wavelength of the electron in this state is $p\pi a_0$ (where a_0 is the Bohr radius). The value of p is

[JEE Advanced-2015]

MP0179

9. In a historical experiment to determine Planck's constant, a metal surface was irradiated with light of different wavelengths. The emitted photoelectron energies were measured by applying a stopping potential. The relevant data for the wavelength (λ) of incident light and the corresponding stopping potential (V_0) are given below: **[JEE Advanced-2016]**

λ (μm)	V_0 (Volt)
0.3	2.0
0.4	1.0
0.5	0.4

Given that $c = 3 \times 10^8$ ms^{-1} and $e = 1.6 \times 10^{-19}$C, Planck's constant (in units of J s) found from such an experiment is :

(A) 6.0×10^{-34} (B) 6.4×10^{-34} (C) 6.6×10^{-34} (D) 6.8×10^{-34}

MP0180

10. Highly excited states for hydrogen like atoms (also called Rydberg states) with nuclear charge Ze are defined by their principal quantum number n, where n >> 1. Which of the following statement(s) is (are) true? **[JEE Advanced-2016]**

(A) Relative change in the radii of two consecutive orbitals does not depend on Z

(B) Relative change in the radii of two consecutive oribitals varies as 1/n

(C) Relative change in the energy of two consecutive orbitals varies as $1/n^3$

(D) Relative change in the angular momenta of two consecutive orbitals varies as 1/n

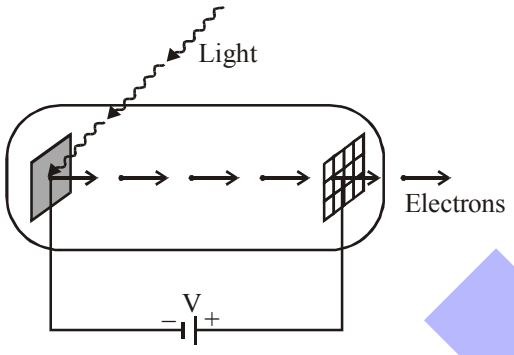
MP0181

11. A hydrogen atom in its ground state is irradiated by light of wavelength 970 Å. Taking $hc/e = 1.237 \times 10^{-6}$ eV m and the ground state energy of hydrogen atom as –13.6 eV, the number of lines present in the emission spectrum is **[JEE Advanced-2016]**

MP0182

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12. Light of wavelength λ_{ph} falls on a cathode plate inside a vacuum tube as shown in the figure. The work function of the cathode surface is ϕ and the anode is a wire mesh of conducting material kept at a distance d from the cathode. A potential difference V is maintained between the electrodes. If the minimum de Broglie wavelength of the electrons passing through the anode is λ_e, which of the following statement(s) is(are) true ? **[JEE Advanced-2016]**



(A) For large potential difference ($V >> \phi/e$), λ_e is approximately halved if V is made four times
(B) λ_e increases at the same rate as λ_{ph} for $\lambda_{ph} < hc/\phi$
(C) λ_e is approximately halved, if d is doubled
(D) λ_e decreases with increase in ϕ and λ_{ph}

MP0183

13. An electron in a hydrogen atom undergoes a transition from an orbit with quantum number n_i to another with quantum number n_f. V_i and V_f are respectively the initial and final potential energies of the electron. If $\frac{V_i}{V_f} = 6.25$, then the smallest possible n_f is. **[JEE Advanced-2017]**

MP0184

14. A photoelectric material having work-function ϕ_0 is illuminated with light of wavelength $\lambda \left(\lambda < \frac{hc}{\phi_0}\right)$.

The fastest photoelectron has a de-Broglie wavelength λ_d. A change in wavelength of the incident light by $\Delta\lambda$ results in a change $\Delta\lambda_d$ in λ_d. Then the ratio $\Delta\lambda_d/\Delta\lambda$ is proportional to

[JEE Advanced-2017]

(A) λ_d^3/λ^2 (B) λ_d^3/λ (C) λ_d^2/λ^2 (D) λ_d/λ

MP0185

15. In a photoelectric experiment a parallel beam of monochromatic light with power of 200 W is incident on a perfectly absorbing cathode of work function 6.25 eV. The frequency of light is just above the threshold frequency so that the photoelectrons are emitted with negligible kinetic energy. Assume that the photoelectron emission efficinecy is 100% A potential difference of 500 V is applied between the cathode and the anode. All the emitted electrons are incident normally on the anode and are absorbed. The anode experiences a force $F = n \times 10^{-4}$ N due to the impact of the electrons. The value of n is..... Mass of the electron $m_e = 9 \times 10^{-31}$ kg and 1.0 eV = 1.6×10^{-19} J.? **[JEE Advanced-2018]**

MP0186

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16. Consider a hydrogen-like ionized atom with atomic number Z with a single electron. In the emission spectrum of this atom, the photon emitted in the n = 2 to n = 1 transition has energy 74.8 eV higher than the photon emitted in the n = 3 to n = 2 transition. The ionization energy of the hydrogen atom is 13.6 eV. The value of Z is..... . **[JEE Advanced-2018]**

MP0187

17. A free hydrogen atom after absorbing a photon of wavelength λ_a gets excited from the state n = 1 to the state n = 4. Immediately after that the electron jumps to n = m state by emitting a photon of wavelength λ_e. Let the change in momentum of atom due to the absorption and the emission are Δp_a and Δp_e, respectively. If $\lambda_a/\lambda_e = \frac{1}{5}$. Which of the option(s) is/are correct ?

[Use hc = 1242 eV nm; 1 nm = 10^{-9} m, h and c are Planck's constant and speed of light, respectively] **[JEE Advanced-2019]**

(1) $\lambda_e = 418$ nm

(2) The ratio of kinetic energy of the electron in the state n = m to the state n = 1 is $\frac{1}{4}$

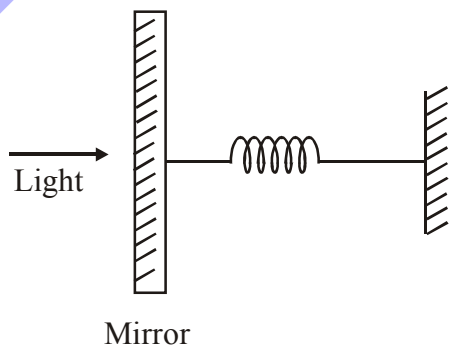
(3) m = 2

(4) $\Delta p_a/\Delta p_e = \frac{1}{2}$

MP0188

18. A perfectly reflecting mirror of mass M mounted on a spring constitutes a spring-mass system of angular frequency Ω such that $\frac{4\pi M\Omega}{h} = 10^{24}m^{-2}$ with h as Planck's constant. N photons of wavelength $\lambda = 8\pi \times 10^{-6}$m strike the mirror simultaneously at normal incidence such that the mirror gets displaced by 1μm. If the value of N is $x \times 10^{12}$, then the value of x is_____.

[Consider the spring as massless] **[JEE Advanced-2019]**



MP0189

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19. In an X-ray tube, electrons emitted from a filament (cathode) carrying current I hit a target (anode) at a distance d from the cathode. The target is kept at a potential V higher than the cathode resulting in emission of continuous and characteristic X-rays. If the filament current I is decreased to $\frac{I}{2}$, the potential difference V is increased to 2V, and the separation distance d is reduced to $\frac{d}{2}$, then

[JEE Advanced-2020]

(A) the cut-off wavelength will reduce to half, and the wavelengths of the characteristic X-rays will remain the same

(B) the cut-off wavelength as well as the wavelengths of the characteristic X-rays will remain the same

(C) the cut-off wavelength will reduce to half, and the intensities of all the X-rays will decrease

(D) the cut-off wavelength will become two times larger, and the intensity of all the X-rays will decrease

MP0378

20. A particle of mass m moves in circular orbits with potential energy V(r) = Fr, where F is a positive constant and r is its distance from the origin. Its energies are calculated using the Bohr model. If the radius of the particle's orbit is denoted by R and its speed and energy are denoted by v and E, respectively, then for the nth orbit (here h is the Planck's constant)- **[JEE Advanced-2021]**

(A) $R \propto n^{1/3}$ and $v \propto n^{2/3}$

(B) $R \propto n^{2/3}$ and $v \propto n^{1/3}$

(C) $E = \frac{3}{2}\left(\frac{n^2h^2F^2}{4\pi^2m}\right)^{1/3}$

(D) $E = 2\left(\frac{n^2h^2F^2}{4\pi^2m}\right)^{1/3}$

MP0395

21. Which of the following statement(s) is(are) correct about the spectrum of hydrogen atom ?

[JEE Advanced-2021]

(A) The ratio of the longest wavelength to the shortest wavelength in Balmer series is 9/5

(B) There is an overlap between the wavelength ranges of Balmer and Paschen series.

(C) The wavelengths of Lyman series are given by $\left(1+\frac{1}{m^2}\right)\lambda_0$, where λ_0 is the shortest wavelength of Lyman series and m is an integer

(D) The wavelength ranges of Lyman and Balmer series do not overlap

MP0396

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22. In a photoemission experiment, the maximum kinetic energies of photoelectrons from metals P, Q and R are E_P, E_Q and E_R, respectively, and they are related by $E_P = 2E_Q = 2E_R$. In this experiment, the same source of monochromatic light is used for metals P and Q while a different source of monochromatic light is used for the metal R. The work functions for metals P, Q and R are 4.0 eV, 4.5eV and 5.5 eV, respectively. The energy of the incident photon used for metal R, in eV, is ___.

[JEE Advanced-2021]

MP0397

23. When light of a given wavelength is incident on a metallic surface, the minimum potential needed to stop the emitted photoelectrons is 6.0 V. This potential drops to 0.6 V if another source with wavelength four times that of the first one and intensity half of the first one is used. What are the wavelength of the first source and the work function of the metal, respectively?

[Take $= \frac{hc}{e} = 1.24 \times 10^{-6}\ Jm\ C^{-1}$.] **[JEE Advanced-2022]**

(A) 1.72×10^{-7} m, 1.20 eV

(B) 1.72×10^{-7} m, 5.60 eV

(C) 3.78×10^{-7} m, 5.60 eV

(D) 3.78×10^{-7} m, 1.20 eV

MP0420

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ANSWER KEY

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	A	B	B	B	C	A	A	D	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	B	A	B	D	D	D	D	A	D	A
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	C	D	C	A	A	A	A	A	A	A
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	D	B	A	B	B	D	B	C	B	A
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	A	A	D	B	C	D	D	C	C	A

EXERCISE # O-2

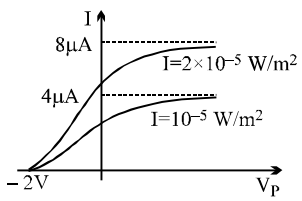
Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	C	A	D	A	B	A	B	D	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	D	B	B	B	D	A,C,D	A,B	B,C,D	B,C
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	A,B,C	A,B,D	A,C	A,B,C	B,C,D	A,D	A,B,C,D	A,B,D	A,C	A,B,C,D
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	B,D	B,D	B,D	A,B,D	B,C	B,C	A,C,D	A,B,C	A,C,D	A,C,D
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	B	A,C	A,C,D	A,B,C,D	A,B,C	A,C,D	A	A	B	A
Que.	51	52	53	54	55	56	57	58	59	60
Ans.	A	B	B	C	B	C	D	A	D	B
Que.	61	62					63			
Ans.	C	(A)-P,R,S; (B)-P,Q; (C)-P,Q; (D)-R,S					(A)-Q,R; (B)-P,S; (C)-Q,S; (D)-P,R			
Que.	64				65	66				
Ans.	(A)-P,R; (B)-Q,S; (C)-Q,S; (D)-P,R				B	(A)-S; (B)-R; (C)-Q; (D)-P				

EXERCISE # S

1. 6.59×10^{-34} Js

2. (a) 3.38×10^{-19} J = 2.11 eV (b) 3.0×10^{20} photons/s

3.



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4. Use $eV_0 = h\nu - \phi_0$ for both sources. From the data on the first source, $\phi_0 = 1.40$ eV. Use the value to obtain for the second suorce $V_0 = 1.50$ V.

5. (a) 3.14×10^{-19} J , 1.05×10^{-27} kg m/s (b) 3×10^{16} photons/s (c) 0.63 m/s

6. 7 **7.** (a) 4.04×10^{-24} kg ms^{-1} (b) 0.164 nm

8. (a) 6.95×10^{-25} J = 4.34 μeV (b) 3.78×10^{-28} J = 0.236 neV

9. (a) 6.686×10^{-21} J = 4.174×10^{-2} eV (b) 0.145 nm **10.** $\sqrt{2}$ **11.** 2.44 **12.** 7

13. (a) $\lambda_n = \frac{2L}{n}$, n = 1,2,3..... (b) $p_n = \frac{nh}{2L}$, n = 1,2,3.... (c) $K_n = \frac{n^2h^2}{8L^2m}$, n = 1, 2, 3,....

14. 4 **15.** 0.55 eV **16.** 7 **17.** 2 **18.** 820 nm.

19. Lyman series: 103 nm and 122 nm; Balmer series: 656 nm.

20. (a) +3.4 eV; (b) –6.8 eV; (c) potential energy **21.** 2.56×10^{-13} m; –2.8 keV

22. 8 **23.** n = 24 **24.** 2

25. Use $\lambda = (hc/E)$ with $E = 5.1 \times 1.602 \times 10^{-10}$ J to get $\lambda = 2.43 \times 10^{-16}$ m.

26. 6210 eV **27.** 400 **28.** 391 eV

EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	4	4	1	4	3	2	3	2	4	4
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	3	4	Allen (Bonus) NTA (1)	1	3	4	4	4	1	3
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	1	3	1	11	2	3	3	4	2	1
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	(10553) or (10553.14)	51.00	2	50.00	4	2	1	25	3	1
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	4	3	B	C	B	D	A	B	114	B
Que.	51	52	53	54	55	56	57	58		
Ans.	3	C	B	B	B	D	D	84		

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	A,C	B	A	2	A,C,D	A,C	2	B	A,B,D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	6	A	5	A	24 [23.60, 24.40]	3 [3,3]	2,3	1.00	A,C	B,C
Que.	21	22	23							
Ans.	A,D	6	A							

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